



Cairo University – Faculty of Engineering
Petrochemical Engineering Program
Credit-hour System

POLYMERIZATION

Petrochemicals from Oil & Gas (CHEN 402)

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Polymers

- You may think of polymers as being a relatively modern invention

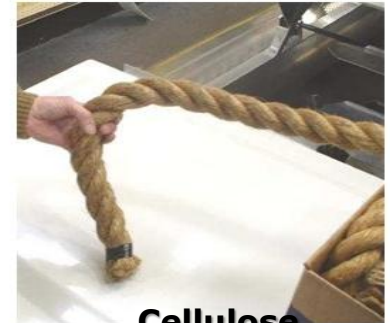
however naturally occurring polymers have been used for thousands of years: wood, rubber, cotton, wool, leather, silk,.. etc



Leather



Wood



Cellulose

- Artificial (Synthetic, man-made) polymers are, indeed, relatively recent and mostly date from after WWII
 - in many cases, the artificial material is both better and cheaper than the natural alternative

Polymers

Made up of chain molecules long-range connections throughout material.

Hydrocarbons

- hydrogen and carbon, bonded covalently
- Simplest are methane, ethane, propane, butane
- C_nH_{2n+2} the paraffins family

In practice, mostly carbon atoms along length of the chain in addition to various side-groups (e.g. -H, -Cl, -OH, -CH₃) attached to it at regular intervals.



Polymers are composed of hydrocarbon basic units (monomers).

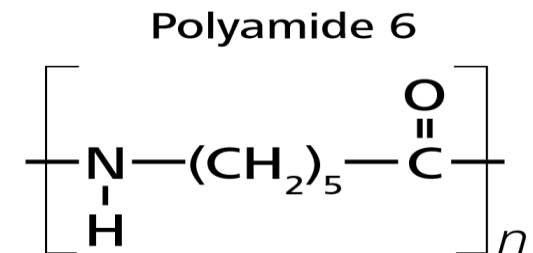
Polymer Materials

High Density Polyethylene (HDPE) for storm water drainage system. Replaces fired clay ceramics (brittle, difficult to make long sections, difficult to join) and cast iron (heavy, low corrosion resistance). Also used in domestic plumbing, displacing copper (expensive).



Kevlar (polyamide – nylon) for body armor. Replaces iron (heavy, stiff).

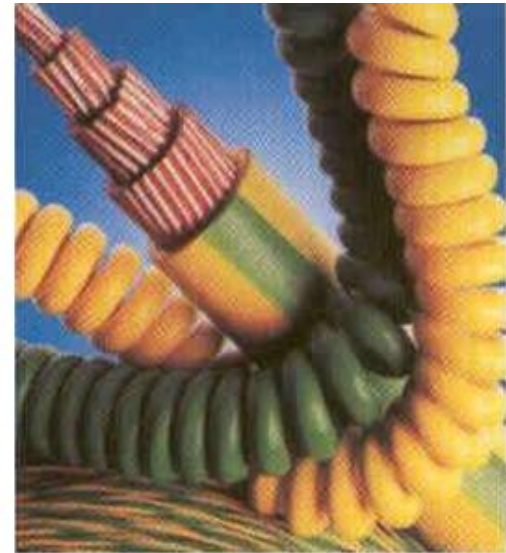
- Bullet proof vest
- Helmet
- Safety boots
- Gloves
- Ropes



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Polymer Materials

Polyvinyl chloride (PVC) for cable insulation: flexible, easily formed, thermally and electrically insulating.



Nylon (polyamide) and polyester for fabrics, replacing silk (expensive) .



Introduction to Polymers



- styro-foam cups
- contact lenses
- rubber tires
- telephone housings
- epoxies
- sandwich bags
- soda bottles
- rubber bands
- computer keyboard
- Keys, cables ... etc

COMMERCIAL Polymers: used in large quantities for their lightweight, corrosion-resistance, and good formability.

- usually low strength and stiffness

ENGINEERING Polymers: improved strength and better elevated temperature properties. Used industrially in machine parts.

In fact, just look around your, house, dorm or apartment room and you'll likely find plenty of examples of polymeric materials.

General Characteristics

- Stiffness is resistance to plastic deformation.
- Strength measures how much stress can be applied to an element before it deforms permanently or fractures.

-Cheap

-Tough (resists elastic deformation)

-Lightweight: **polyethylene density = 0.96 g/ cc**

-Easy to shape

-Flexible

-Recyclable

-Resistant to chemical attack

-Thermally and electrically insulating

BUT

- Low stiffness / strength (does not resist deformation):
 E (polyethylene) = 1 GPa
- Limited temperature range: **T_m (polyethylene) = 105°C**
- High coefficient of thermal expansion:
 α (polyethylene) = 150 x 10⁻⁶ K⁻¹
- Degrade when exposed to UV light, oxygen.

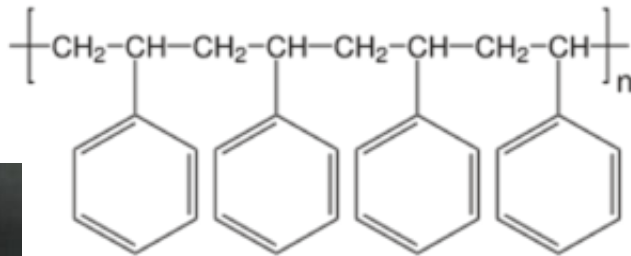


Different Aspects of Polymer Science



- **Polymer synthesis**
- **Polymer characterization**
- **Polymer physical properties**

Each of these aspects has considerable industrial and research significance. The successful polymer scientist has some understanding of all of these areas.



Polystyrene Structure

- **Polymer:** long molecule made up by the repetition of small units (monomer)

Synonym: macromolecule

High molecular weight (M)

Ex. ↗ polystyrene (PS)

- **Monomer:** molecule that combines with others (identical or different) to form a polymer

Ex. ↗ ethylene (PE - polyethylene)

- **Oligomer:** low molecular weight polymer, constituted of at least two monomers



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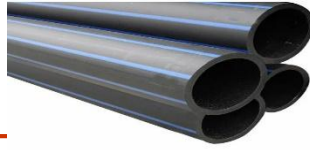
Polymer Material

HOMOPOLYMER

- The polymer molecule is made up from simple units called monomers. The units that are repeated throughout the polymer chain and that characterize the chemical composition of the polymer are known as repeat units.
- The repeat unit is generally composed of 1 monomer unit and thus called homo-polymer or 2 or even more monomer units (called copolymer).
- The polymer consists of mixture of polymer molecules (chains of varying length) of varying molecular weights (homologues) ranging from thousands to millions.
- The pendant groups are those parts of a polymer molecule attached to the main backbone.

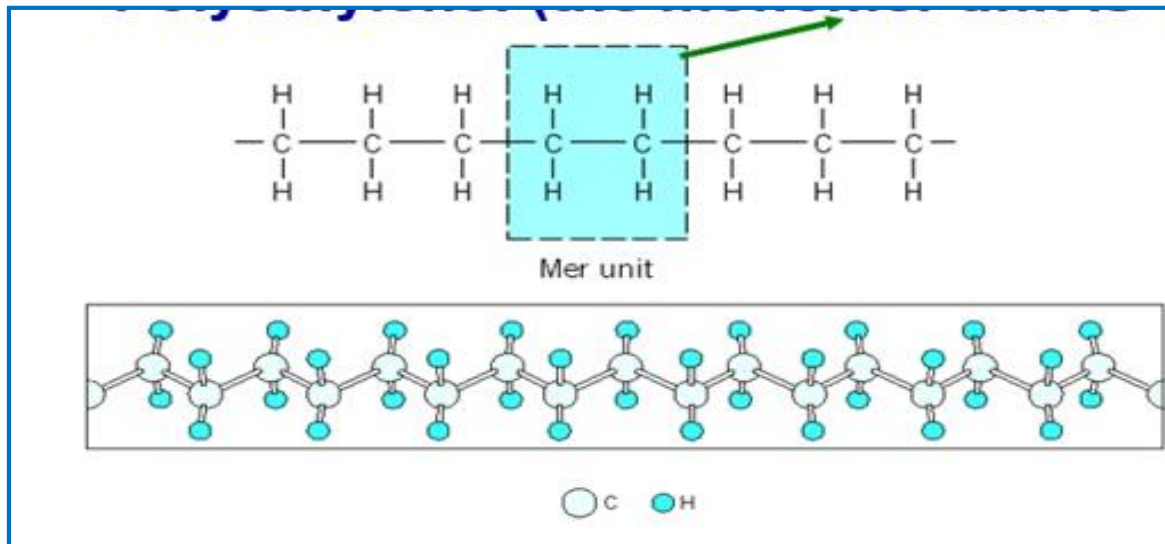
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Examples of Polymers



Examples of “real” monomers and their resulting polymers:

Polyethylene: (the monomer unit is C_2H_4)



③ Polyethylene is used for flexible bottles, toys, battery parts, ice trays, film wrapping and other packaging materials.

③ It is tough but possesses low strength.

③ Trade names: Ethron, Fortiflex, Hi-fax, Rigidex, Zendel.

This is an example of a linear chain homopolymer, where the “X” in our model is replaced with the ethylene group.

Examples of Polymers

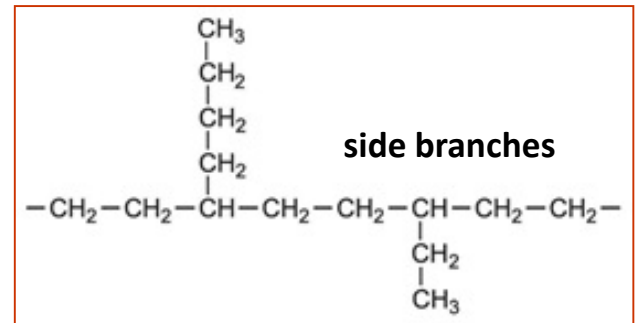
The type of polyethylene that goes into milk/juice/liquids jugs is stronger, stiffer, but more opaque (less optically clear) than the type of polyethylene that is used to make film wrap (greater optical clarity, more flexible, but less strong)



Can you figure out which type of polyethylene is used to make film wrap?

A. High density (No side branches)

B. Low density (Side branches)





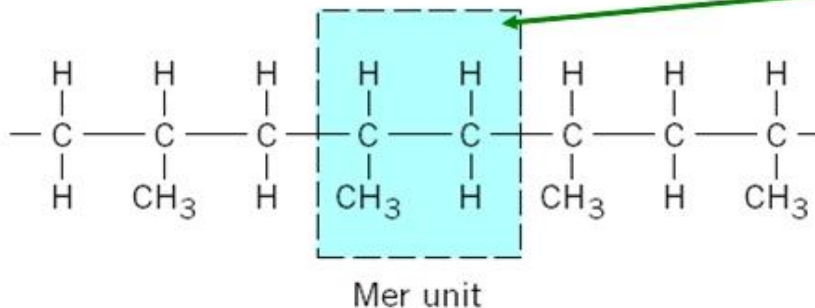
Examples of Polymers



Polypropylene: (the monomer unit is C_3H_6)



propylene

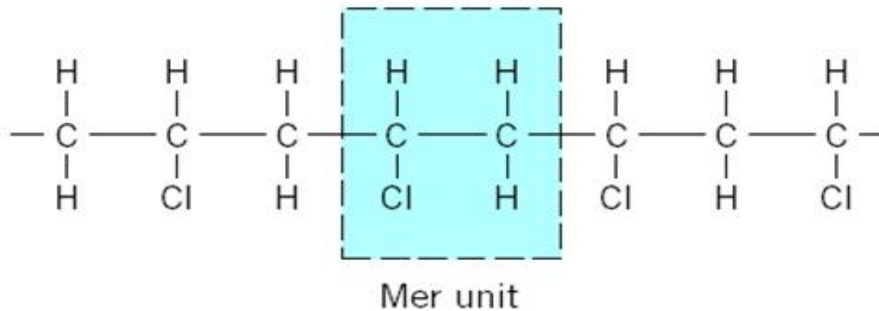


- Polypropylene is used for such items as bottles, TV cabinets, luggage, fibers & fabrics.
- It tends to be relatively strong and resistant to heat.

Examples of Polymers



Polyvinyl chloride: (the mer unit is C_2H_3Cl)



PVC Wall Panels

- Polyvinyl chloride is a very popular, low cost rigid material
- It can be made flexible by adding plasticizers.
- It is used as floor coverings, pipe, garden hose, electrical wire insulation, and (at one time) phonograph records.
- Tradenames: “PVC,” Saran, Tygon, Darvic, Geon.

Factors Affecting Polymers Properties

- Polymers are tailored man-made materials.
- Polymer properties are determined by the following factors:
 - Chain structure (Linearity - Recurrence of the pendant group - Recurrence of repeat unit in copolymerization)
 - Degree of crystallinity
 - Thermal Transitions
 - Stereo-isomerism (Tacticity in polymers)
 - Degree of Polymerization (Molecular Weight)
 - Synthesis techniques, additives, processing

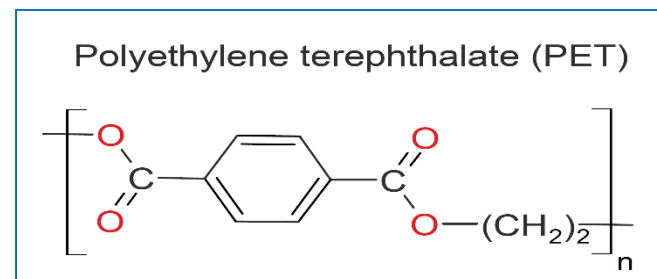
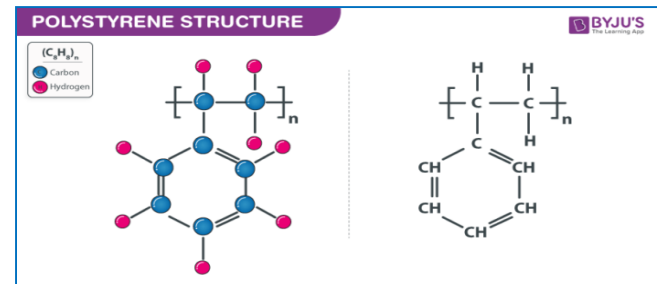
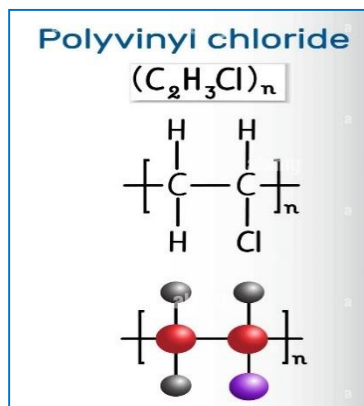
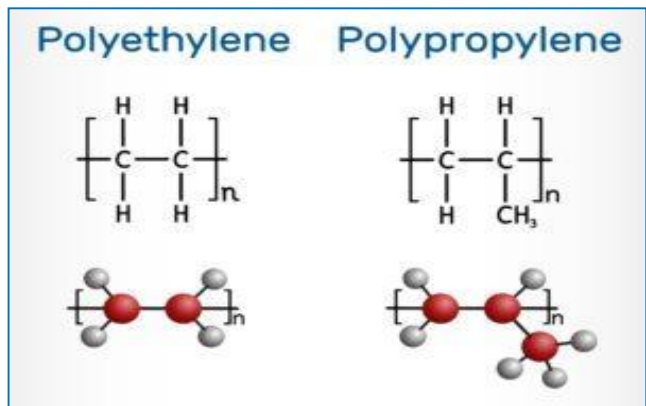
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Classification of Polymers

According to Thermal Properties polymers are either:

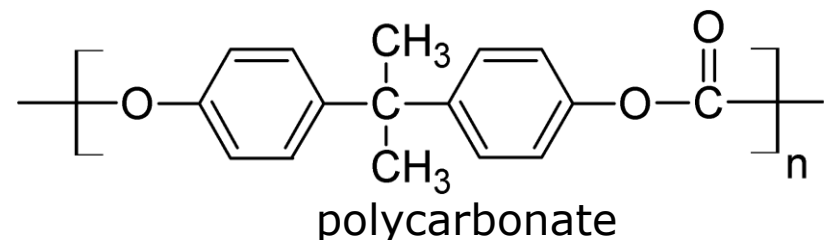
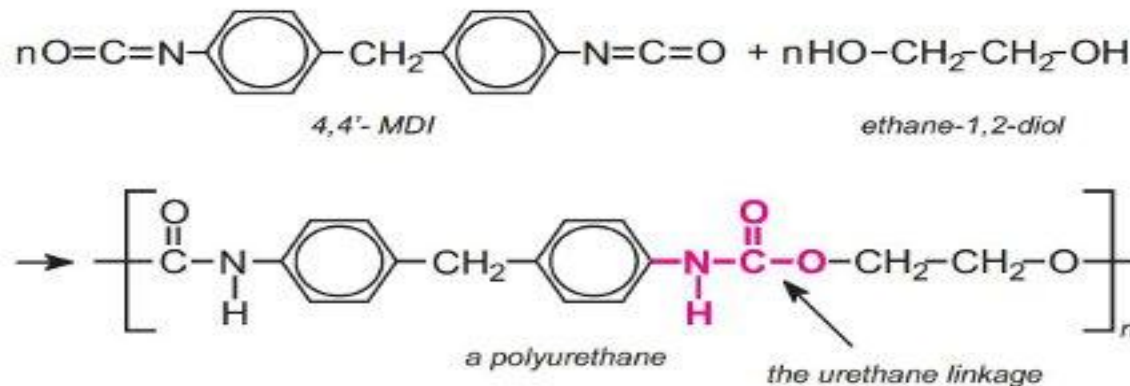
THERMOPLASTICS, THERMOSETS AND ELASTOMERS

- **Thermoplastics**: can be processed by melting (several cycles of heating and cooling are possible for thermally stabilized polymers) such as PE, PS, PP, PVC and PET.



Classification of Polymers

- **Thermosets**: Polymers that cure irreversibly; cannot be melted or dissolved to be processed; **chemical decomposition occurs before softening**, mostly of complex structure and high T_g, such as polycarbonate, polyurethane, vulcanized rubber, etc.



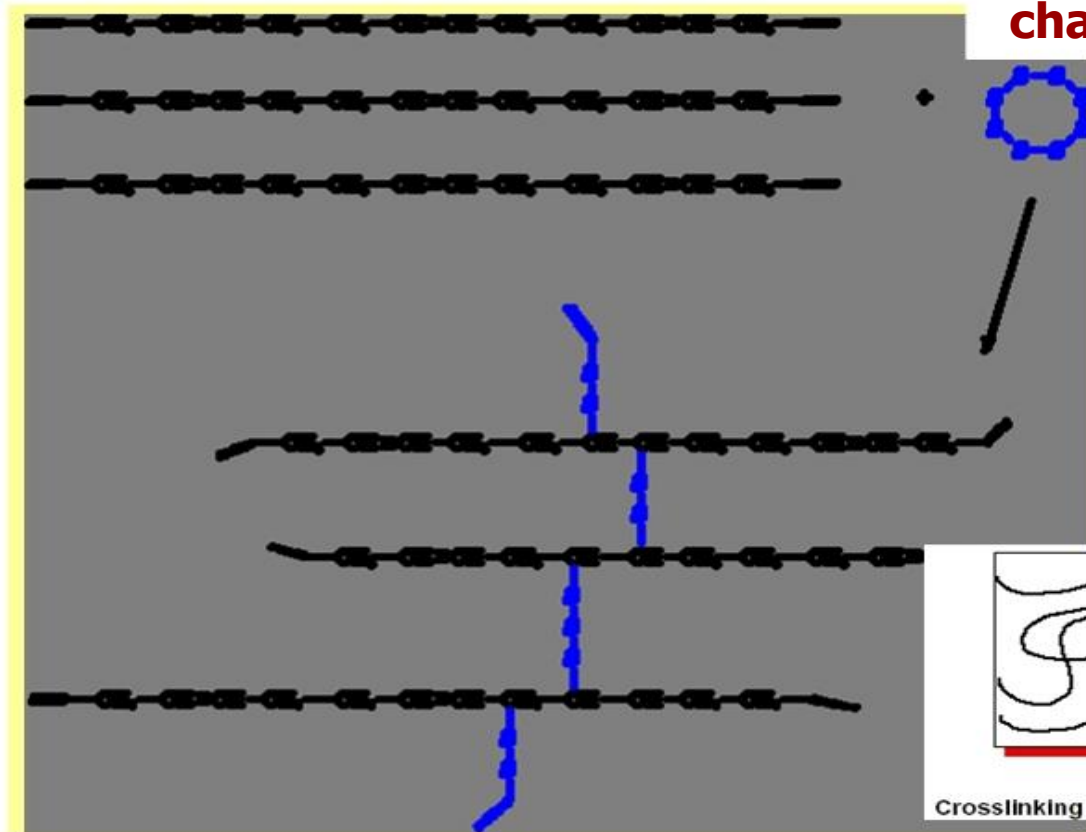
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Vulcanization of Rubber

Difference between thermosets and elastomers:

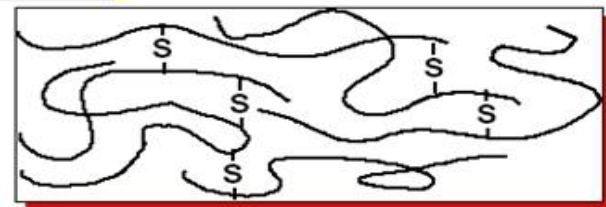
Both decompose on heating; however, elastomers can stretch and return to their original shape due to elasticity while as, thermosets harden irreversibly with heat or chemical curing.

ELASTOMERS



Sulphur reacts with chain at the double bond

This kind of materials are referred to as elastomers



Crosslinking Prevents One Chain From Sliding Over Another

Basic Polymer Structure

Linear Polymers

Suppose our repeat unit is “X.” Then, a linear polymer based on “X” would look like the following:

... --X--X--X--X--X--X--X--X--X--X--X--X--...

where each “X” represents a “monomer.”

Sometimes, polymers contain functional side groups, called pendant groups:



We call the primary linear chain the “backbone”

Molecular Structure

According to their Chain Structure, polymers are either:

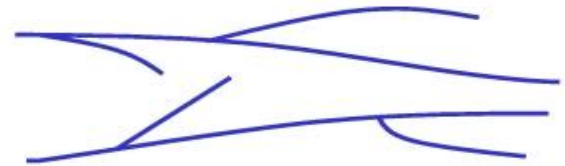
- **Linear polymers**

- long, flexible chains (with only 2 ends) with some van der Waals or hydrogen bonding between chains



- **Branched polymers**

Chain packing efficiency is reduced compared to linear polymers (lower density)



- **Crosslinked polymers**

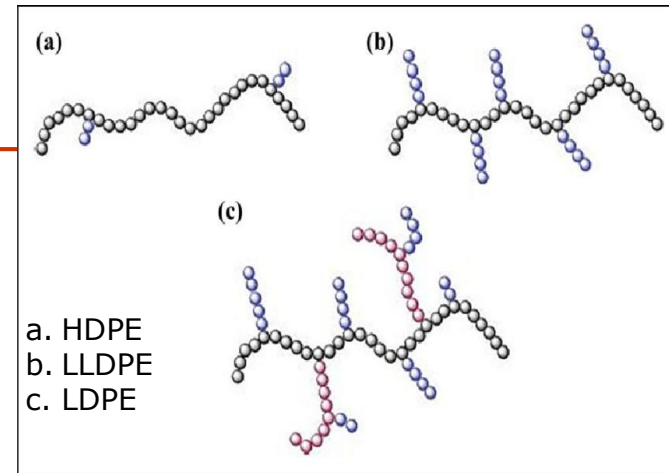
- cross linkage happens either during synthesis or in a separate process, typically involving addition of impurities which bond covalently
- this is termed vulcanisation in rubber



- The same polymer may be synthesized in each of the above mentioned types by applying different catalysts and operating conditions.
- SIDPEC in Alexandria Egypt produces the linear (HDPE) and the branched LDPE.

Effect of Cross Linking

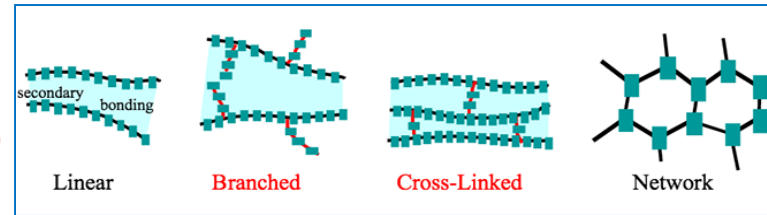
- **Linear and branched polymers are thermoplastic.**
 - Their molecules are free to slide.
 - They can be dissolved in suitable solvents.
- **Cross-linked polymers are thermosetting.**
 - Their molecules are not free to flow or dissolve.
 - Cross links can be imparted during polymerization or in post polymerization such as in vulcanization of rubber and/or by using a hardener.
 - Cross linked polymers differ from each other due to
 - The character of the main chain.
 - The character of the cross links.
 - The degree of cross linking.



Comparison between different types of PE

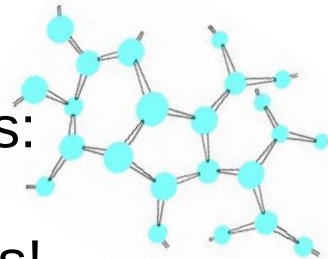
HDPE	LLDPE	LDPE
High density ($\rho > 940$ Kg/m ³)	Density between HDPE and LDPE (910 – 940 Kg/m ³)	Low density ($\rho = 910 - 925$ Kg/m ³)
High crystallinity	Between HDPE and LDPE	Low crystallinity
High melting point	Between HDPE and LDPE	Low melting point
Low MW	Between HDPE and LDPE	High MW
High mechanical properties	Between HDPE and LDPE	Low mechanical properties
No branches at all	Short and arranged branches	Many long unarranged branches
Use Ziegler or Phillips catalyst and no additives	Use Ziegler or Phillips catalyst	Use free radicle catalyst
25 bar and less than 100oC	20 bar and less than 100oC	More than 2000 bar and 300oC.
Used in applications requiring strength, rigidity, and chemical resistance, such as bottles, pipes, containers and industrial equipment	Used in films and packaging due to its flexibility and toughness	Used for applications requiring flexibility, such as plastic bags, films, and coatings
Arranged in a simple, straight shape, without branches or curves provides superior tensile strength and durability.	Due to its higher crystallinity than LDPE it has better chemical resistance, stiffness, and strength than LDPE.	Its highly branched structure gives it excellent flexibility and impact resistance.

Molecular Structure

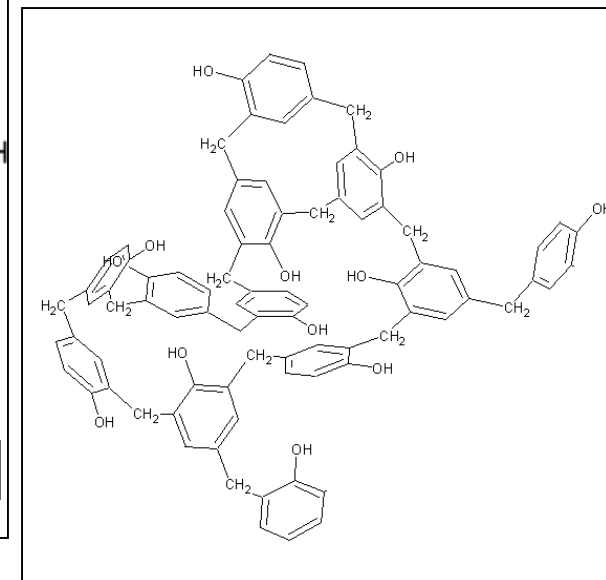
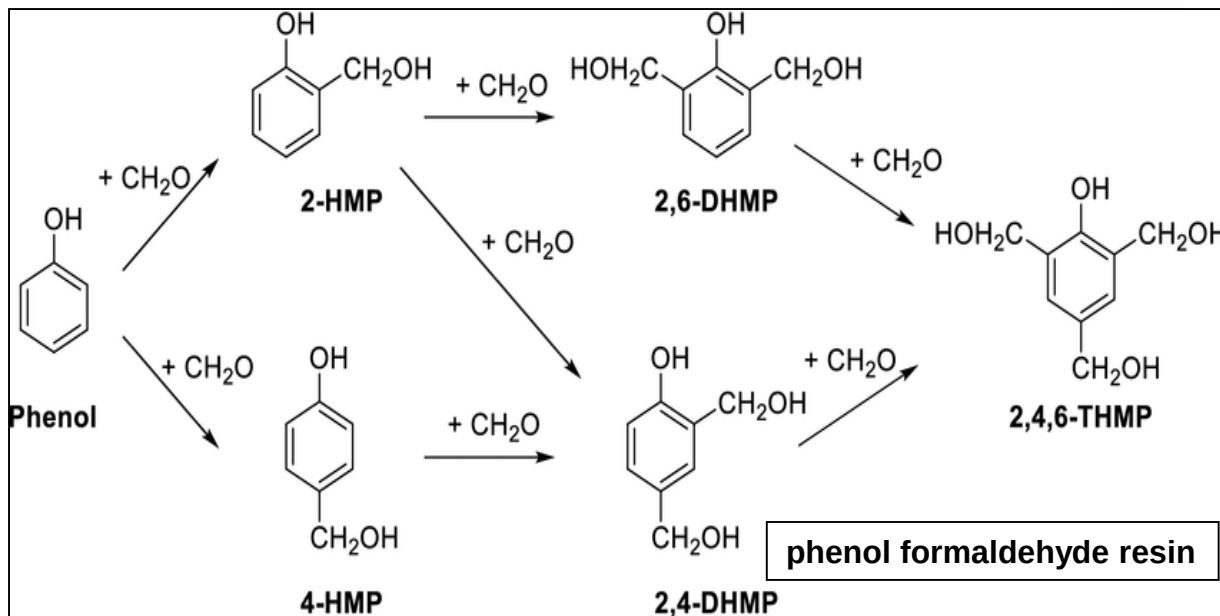
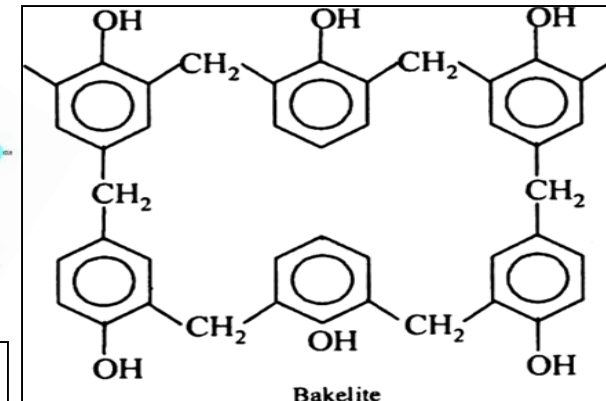


Network polymers:

3D networks made from trifunctional monomers. Examples: epoxies, phenol formaldehyde. These are thermosetting materials!



Phenol formaldehyde Resin



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Homo-polymers & Co-polymers

According to their backbone, polymers are either:

Homopolymers vs. co-polymers:

-If only one type of repeat unit is present, the polymer is called a **homopolymer**.

-If a second monomer is also present in the chain, the resulting material is called a **co-polymer**.

- **Copolymerization:** It is the simultaneous polymerization of 2 or more monomer systems, each of which is capable of polymerizing by it- self (VI).
- In a sense copolymerization does for high polymers what alloying does for metals. Of the synthetic polymers used today, by far the greater number are copolymers, we can define three types of copolymers, when a 2 monomers system is used.

Homo-polymers & Co-polymers

In block copolymers, each monomer undergoes polymerization in a separate reactor where the length of the chains is controlled to a specific size then they are copolymerized in a third reactor.

- ❖ Model for a homo-polymer:

--X--X--X--X--X--X--X--X--X--X--X—X

- ❖ Model for an alternating co-polymer: The units appear in the chain in alternating positions. This regular structure requires using special catalysts.

X--W--X--W--X--W--X--W--X--W--X—W

- ❖ Model for a random co-polymer: The units X and W occur randomly along the polymer chain leading to an irregular configuration.

--X--X--X--W--X--W--X--X--W--X--W—W

- ❖ Model for a block co-polymer: The units appear in long alternating segments of random length.

--X--X--X--X--X--X--X--X--W--W--W--W--W--W--

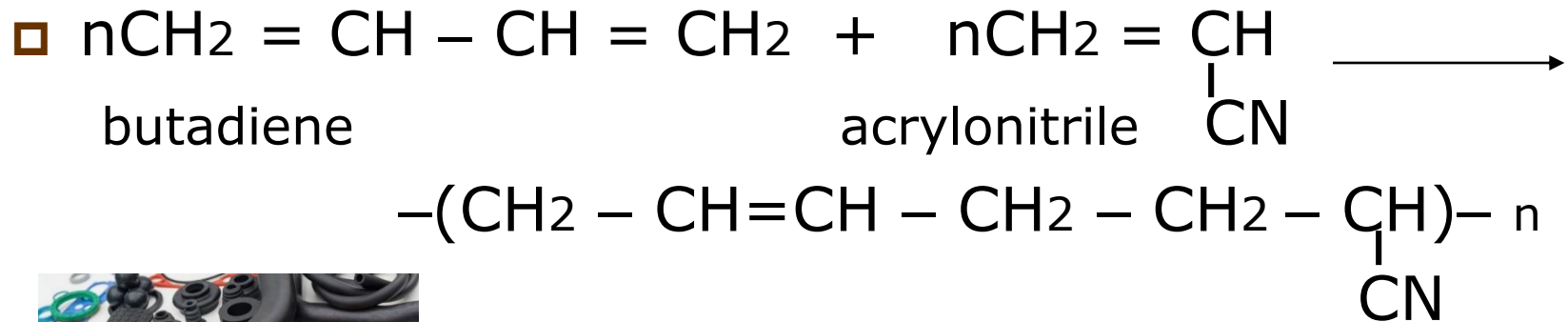
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Copolymerization

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- In a sense copolymerization does for high polymers what alloying does for metals.**
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Examples of Copolymers



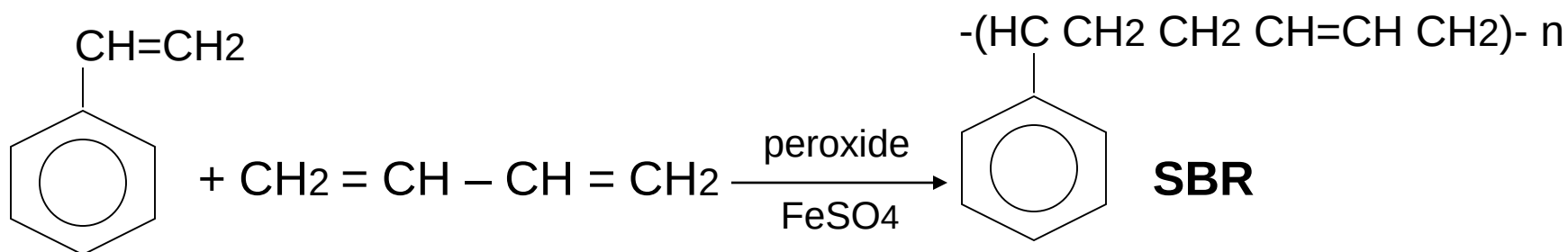
butadiene-nitrile rubber

Copolymerization of butadiene and acrylonitrile results in a butadiene-nitrile rubber which possesses good oil and petrol resistance and hence it is used in manufacturing hose and gaskets for automobiles and other engines.

Styrene-butadiene Rubber (SBR)



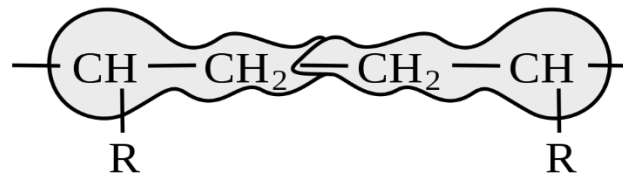
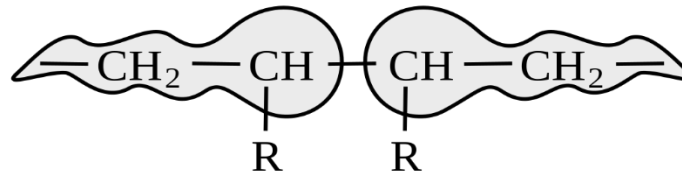
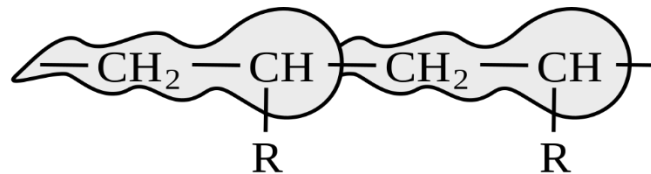
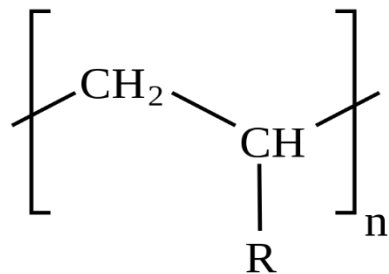
- Copolymerisation of styrene (25%) and butadiene (75%) gives a synthetic rubber known as SBR.



SBR is the most important synthetic rubber used by the tire industry, due to its good mechanical and physical properties. SBR is usually reinforced with carbon black to improve its abrasion, resilience and ageing characteristics.

Reccurance of the Pendant Group

- It is the regularity with which a certain group appears along the chain. It is due to the different ways of monomer combination leading to different polymer properties. The configurations below gives regular polymer, while as combination of them gives irregular polymer.

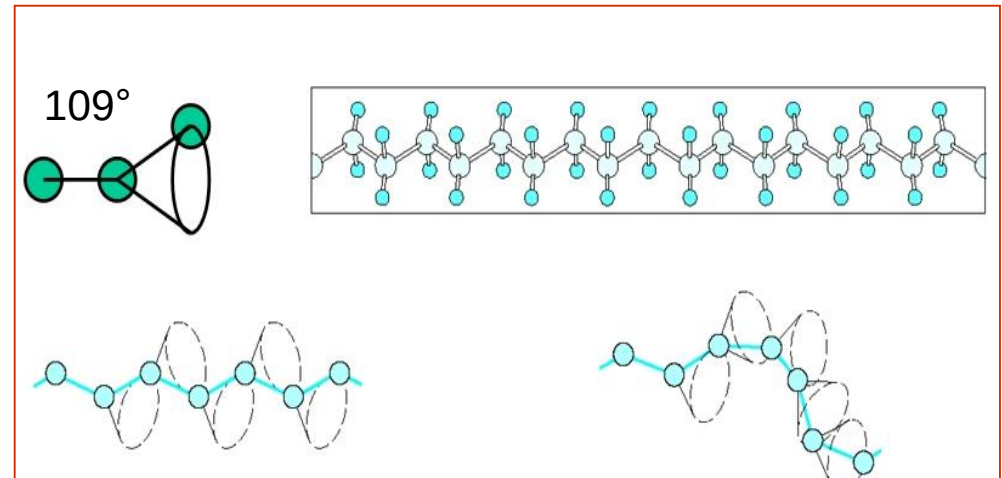
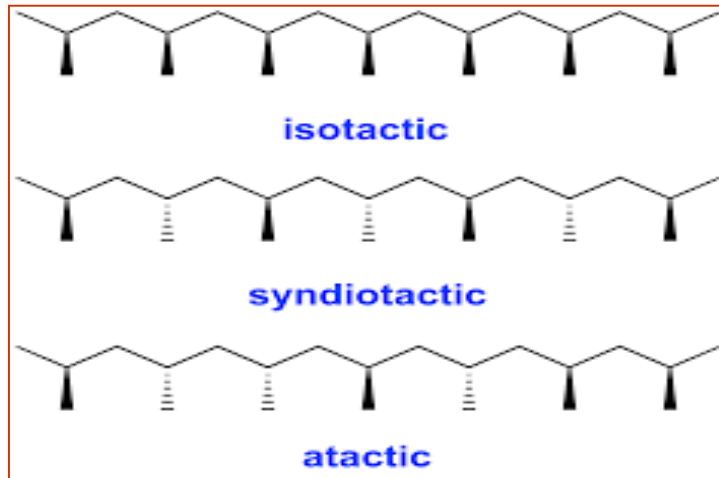


Stereo-isomerism (Tacticity)

According to arrangement of atoms in space, polymers are either:

Isotactic, Syndiotactic or atactic

- If the form of the molecule was strictly determined, polymers would be straight. Steric hindrance hinders atoms from straight configuration leading to helix or zigzag like configuration.
- In fact, the 109° bond angle in polyethylene gives a cone of rotation around which the bond lies.

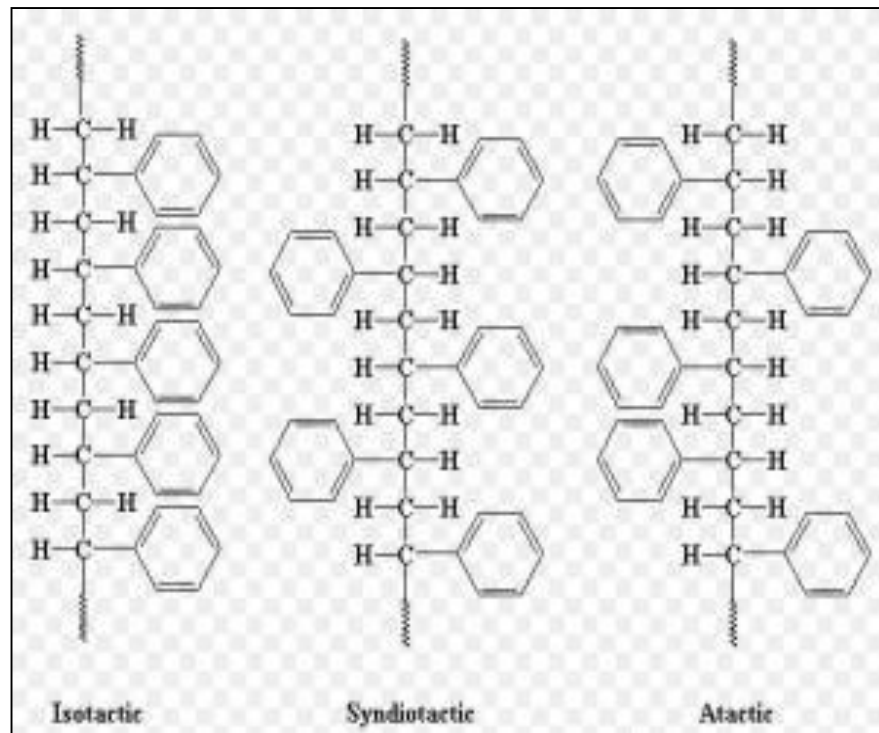


Classification of Polymers Stereo-Isomerism

(Tacticity in Polymers)

If the polymer chain is drawn in a zig-zag fashion, each of the substituent groups will necessarily be located above or below the plane defined by the carbon chain.

1. If all the substituents lie on one side of the chain the configuration is called **isotactic**.
2. If the substituents alternate from one side to another in a regular manner the configuration is termed **syndiotactic**.
3. Finally, a random arrangement of substituent groups is referred to as **atactic**.

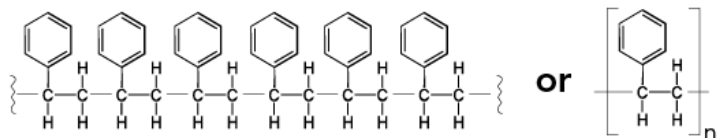


Polystyrene

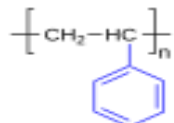
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Stereo-Isomerism of Polystyrene

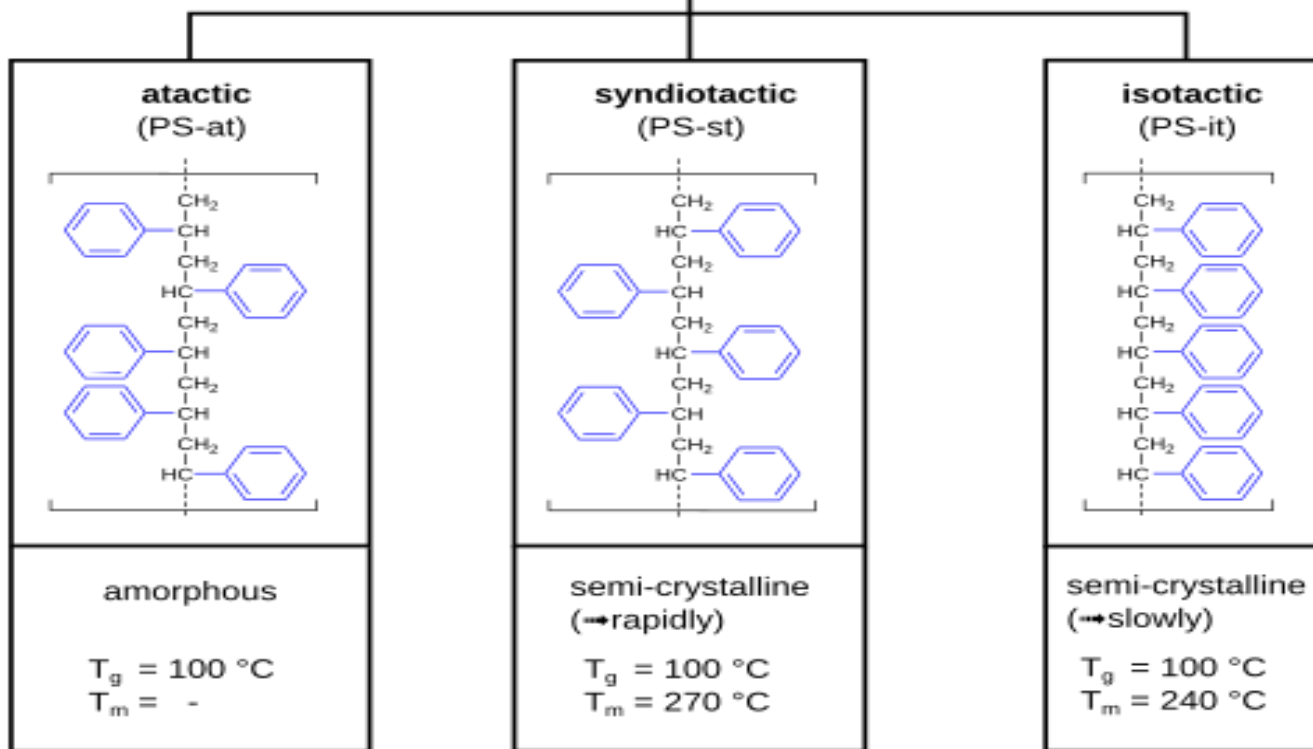
Polystyrene



Polystyrene (PS)



Isotactic & syndiotactic polymers are regular leading to an opaque high performance polymer with high crystallinity, MP and mechanical properties.



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Classification of Polymers

Stereo-Isomerism

- Many common and useful polymers, such as polystyrene, polyacrylonitrile and polyvinyl chloride are atactic as prepared.
- Normally polymers are produced as atactic (irregular).
- Using specific catalysts such as Zeigler-Nata catalyst produces regular isotactic polypropylene.
- The properties of a given polymer will vary considerably with its tacticity. Thus, atactic polypropylene (MP ca. 130°C) is useless as a solid construction material, and is employed mainly as a component of adhesives or as a soft matrix for composite materials.
- In contrast, isotactic polypropylene is a high-melting solid (ca. 170 °C) which can be molded or machined into structural components.

Polymer	T _g atactic	T _g isotactic	T _g syndiotactic
PP	-20 °C	0 °C	-8 °C

Stereospecific Polymerization

- **Stereospecific polymerization is a method of tailoring atomic arrangements within chains to produce oriented polymers with unusual properties.**
- **These properties include: Higher density, higher crystallinity, higher melting points and improved mechanical properties.**
- **Stereospecific agents:**
 - **Ziegler catalysts.**
 - **Supported metal oxide catalysts.**
 - **Radiation (X and γ rays)**

Molecular Forces in Polymers

- **Intramolecular forces:** generally covalent bonds (strong) (forces between atoms in one chain)
- **Intermolecular forces:** Van der Waals forces (PE), hydrogen bridges (PS) (stronger). Can become very strong as $M \uparrow$ (forces between two chains)
- **Entanglements** (physical)



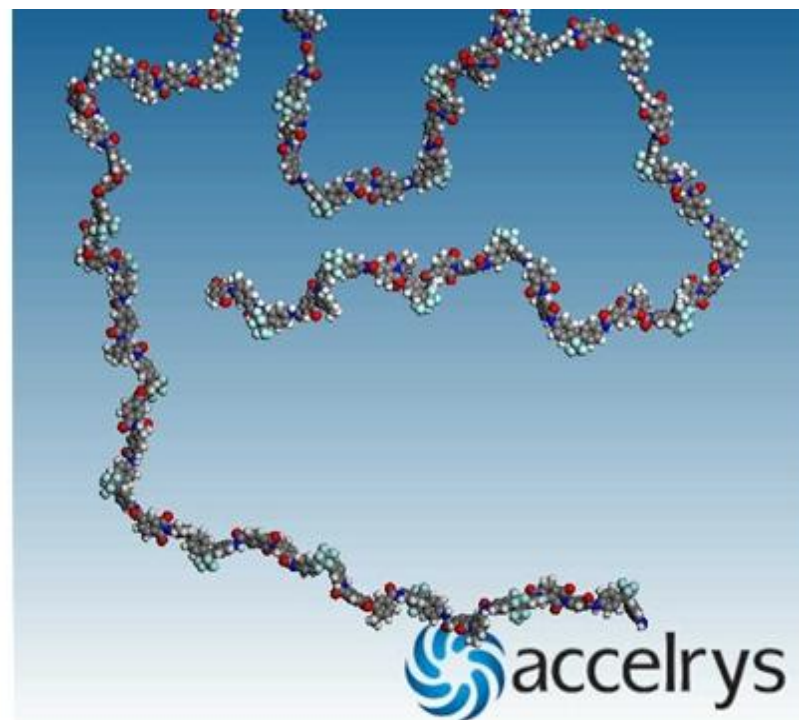
In the crystalline state the Van der Waals bonds are very important.

In the rubbery amorphous state the entanglements are very important.

Molecular Shape

- Hence the polymer chain can bend, twist, and kink into many shapes
- *Adjacent molecules can intertwine leading to the highly elastic nature of many polymers, such as rubber*

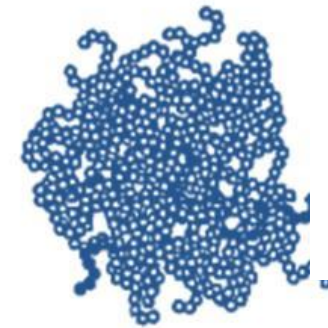
Random kinks and coils lead to entanglement, like in the spaghetti structure



Polymers Classifications

Morphology (Physical State) (1)

1) **Amorphous:** Polymer whose molecular structure lacks a definite repeating form, shape, or structure. Below the glass transition temperature (T_g), the polymer is glassy. Above the T_g molecules are considered liquids that may flow (however, high viscosity may hinder the flow). Amorphous polymers do not have a melting point (T_m), but they soften over a temperature range



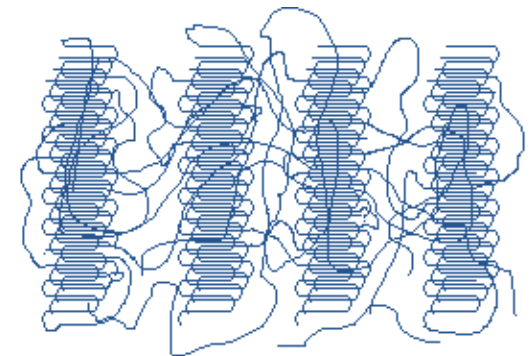
Random coil
"cooked
spaghetti"



Crystalline
"uncooked
spaghetti"

2) **Crystalline:** Polymer whose molecules pack together in ordered regions called crystallites. Pure (100%) crystalline polymers are rarely encountered (very difficult. Why?).

Usually the polymer exhibits two types of morphology in the solid state: amorphous and crystalline. So, the appropriate name for the crystalline polymers is: semi-crystalline polymers



Semi-crystalline

Mechanism of Crystallinity in Polymers

Pure (100%) crystalline polymers are rarely encountered (very difficult. Why?)

- Polymers are composed of long molecular chains which form irregular, entangled coils in the melt.
- Some polymers retain such a disordered structure upon freezing and thus convert into amorphous solids.
- In other polymers, the chains rearrange upon freezing and form partly ordered regions.
- Although it would be energetically favorable for the polymer chains to align parallel, such alignment is hindered by the entanglement.
- Therefore, within the ordered regions, the polymer chains are both aligned and folded.
- Those regions are therefore neither crystalline nor amorphous and are classified as semi-crystalline.

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Mechanism of Crystallinity in Polymers

Crystallinity depends on the molecular structure. WHY?

- Examples of semi-crystalline polymers are linear polyethylene (PE), polyethylene terephthalate (PET), or isotactic polypropylene (PP).
- Whether or not polymers can crystallize depends on their molecular structure – **presence of straight chains with regularly spaced side groups facilitates crystallization.**
- For example, **crystallization occurs very much easier in isotactic and syndiotactic PP than in the atactic PP form.**
- **Atactic polymers crystallize when the side groups are very small**, as in polyvinyl chloride and do not crystallize in case of large substituents like in rubber or silicones.

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Effect of Crystallinity on Polymer Properties

Crystalline polymers possess high density, high thermal conductivity and high MP

Property	Change with Increasing Degree of Crystallinity	
Strength Ability to withstand stress	Generally increases with degree of crystallinity	Crystallinity decreases with increasing branching. Branches prevent the polymeric chains from aligning to form crystalline domains
Stiffness Resistance to deformation	Generally increases with degree of crystallinity	
Toughness Ability to deform plastically without fracturing	Generally increases with degree of crystallinity	
Barrier Properties	Small molecules usually cannot penetrate or diffuse the crystalline domains, hence “barrier properties”, increase with degree of crystallinity	
Solubility	Similarly, solvent molecules cannot penetrate the crystalline domains leading to decreased solubility. So, generally, solvent resistance increases with degree of crystallinity	
Optical Clarity Transparency	Generally decreases with increasing degree of crystallinity. Semi-crystalline polymers usually appear opaque because of the difference in refractive index of the amorphous and crystalline domains, which leads to scattering (will depend upon crystallite size)	

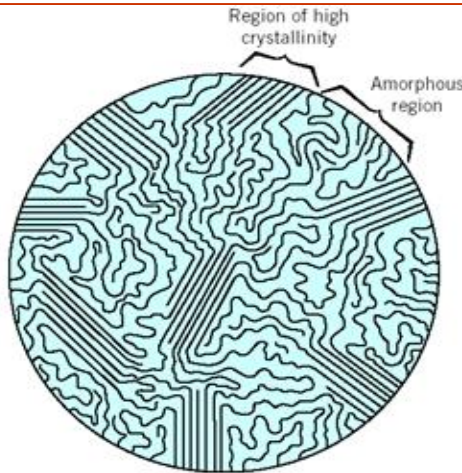
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Crystallinity in Polymers

What are the Factors that favor crystallizing

1. **Structure:** The extent to which a polymer crystallizes depends on whether its structure is encouraging to packing into the crystalline state or not. Packing is facilitated for polymer chains that have:
 - 1) Structural regularity (isotactic and syndiotactic versus atactic),
 - 2) Compactness (no or small pendant groups),
 - 3) Streamlining (no or low branching),
 - 4) Some degree of flexibility (but not so much flexibility as this has an adverse effect on packing)
2. **Secondary attractive forces:** The stronger the secondary attractive forces such as hydrogen bonds, the greater will be the driving force for the ordering and crystallization of polymer chains.

Crystallinity in Polymers



Note the difference between the two terms: Crystalline polymer and crystallizable polymer.

Most real polymers contain both amorphous and crystalline regions, as shown here.

% Crystallinity depends on several factors:

- Rate of cooling (faster cooling - less crystallinity)
- Type of polymer (simple structures - more crystallinity)
(Copolymers - less crystallinity)
- Linear polymers form crystals more easily

-Degree of Crystallinity ranges from 5 - 95%

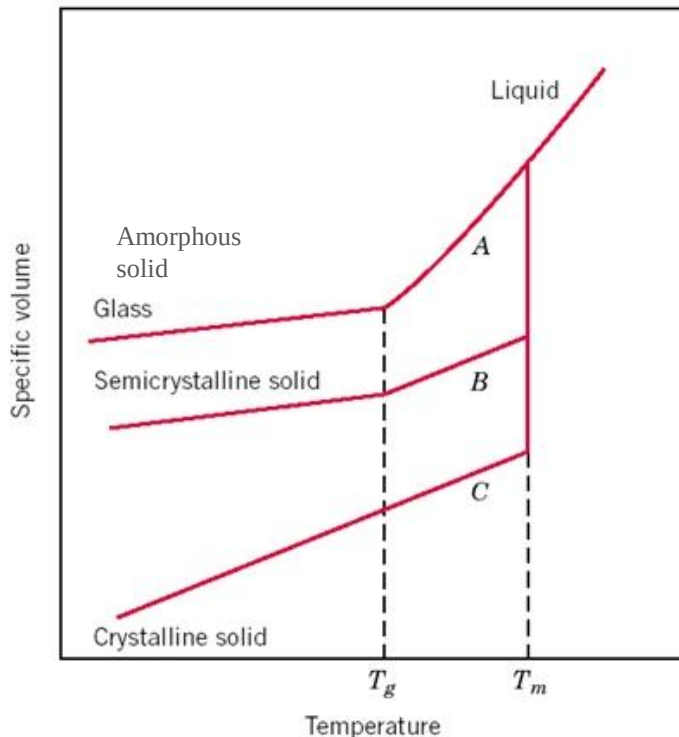
-The higher % Crystallinity → higher strength

Thermal Transitions in Polymers

Melting and Glass Transition Temperature

For amorphous and semi-crystalline polymers, this is a critical aspect of designing polymers.

- Polymers melt over a range of temperatures due to variation in the length of the molecules.
- On identifying the T_m of a certain polymer we consider the highest temp. range. WHY?!?



- **Crystalline polymer:** there is a discontinuous change in specific volume at T_m
- **Amorphous polymer:** continuous, no T_m - but there is an increase in slope at T_g , the glass transition temperature,
- **Semicrystalline polymer:** intermediate to crystalline and amorphous, show both transitions.

- Glass transition temperature is the temperature at which an amorphous polymer changes from a hard/glassy state to a soft/rubbery state, or vice versa.
- T_g is directly related to a material's strength and capabilities in any given end-use application.

Thermal Transitions in Polymers

Melting point T_m :

It is the melting temp. of crystalline part of semi-crystalline polymer. It differs than that of low molecular weight crystalline solids in the following:

- 1- It is a range $(2-10)^\circ\text{C}$
- 2- T_m is subject to strong hysteresis.
- 3- On melting, polymers become viscous liquids.

Hysteresis in state transitions is when melting temperature and freezing temperature do not agree.

Example: agar melts at 85°C and solidifies at 40°C

Glass transition temperature T_g :

- It is associated with amorphous part of polymers.
- At T_g the properties change such as:
 - Thermal conductivity, specific heat, shear modulus and dielectric constant and refractive index change.
 - T_g can be changed by addition of solvents, plasticizers, M.W. distribution and many other additives.

Thermal Transitions in Polymers

- **Melting of a crystalline polymer**

- transforming solid with an ordered structure to a viscous liquid with a highly random structure

- **Amorphous glass transitions**

- transformation from a rigid material to one that has rubber- like characteristics
 - temperature has large effect on chain flexibility

- **Below glass transition temperature, T_g ,**

polymers are usually brittle and glass-like in mechanical behavior.

- **Above glass transition, T_g ,**

polymers are usually more elastic.

Why?



Bond rotations are “freezing” which means chains can’t slip past each other so polymer becomes brittle, (no plastic deformation)

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Polymer Behavior Above and Below T_g

Below T_g:

Polymer chains are “frozen” into place, only bond stretching possible.

Polymer is “glassy”: stiff, strong and brittle

Above T_g:

Polymer chains are free to move around and change shape

Elastomers: chain motion restricted by cross-linking

Semi-crystalline polymers: amorphous sections pinned by crystalline regions

Amorphous polymers: slight restriction on chain motion provided by entanglements, but resistance to deformation drops rapidly with temperature, until material flows. Amorphous polymers are generally used below T_g

Factors Influencing T_g

- Chain flexibility
- Cooling rate
- Side Group size
- Cross-linking
- Branching
- Chain length

Typical glass transition temperatures:

Polyethylene: -90°C

Polypropylene: -18°C

Polystyrene: +100°C

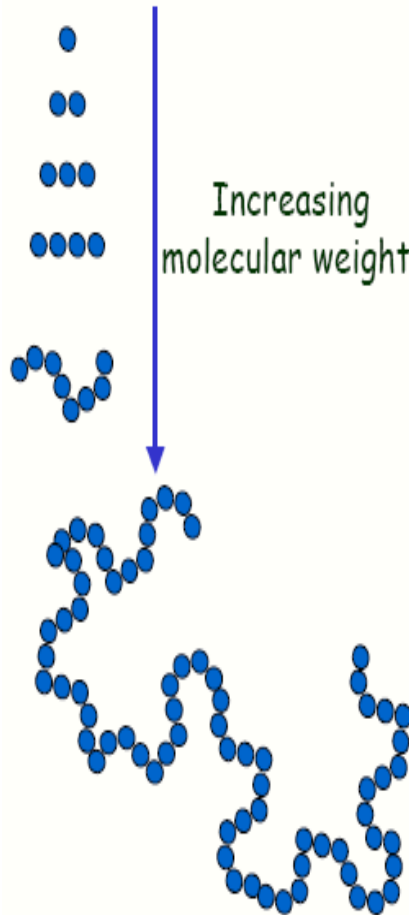
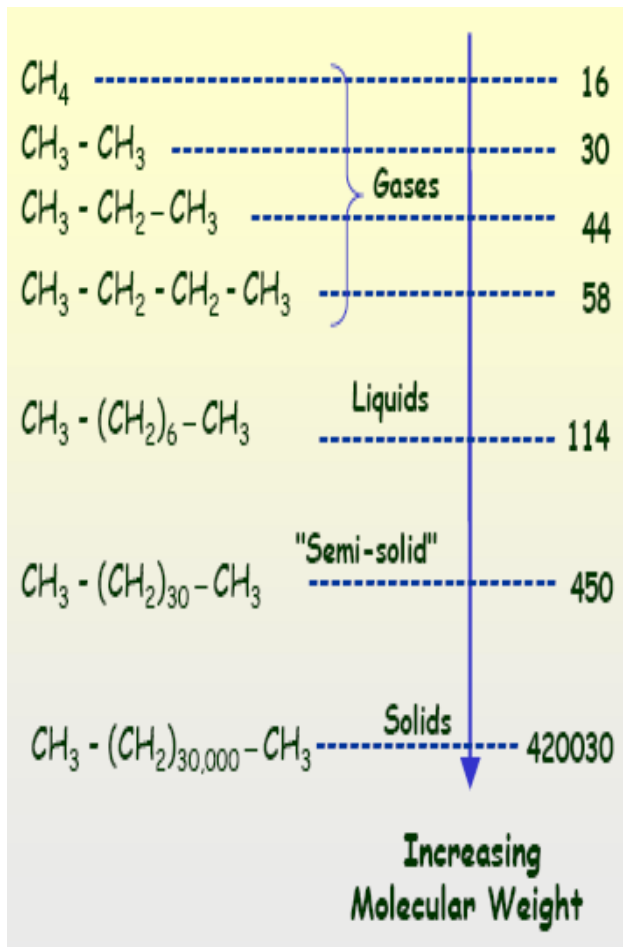
Polymers and Spaghetti

- Amorphous polymers....hot fresh spaghetti with no “clumps”
- Semicrystalline polymers....hot fresh spaghetti with some “clumps”
- Crystalline polymers....spaghetti the way my friend made it, mostly “clumps” with some free strands
- ...Glassy...Polymer below T_g , three day old spaghetti- left in the sun!

	<i>Material</i>	<i>Glass Transition Temperature</i> [°C (°F)]	<i>Melting Temperature</i> [°C (°F)]
T_g is low for simple linear polymers	Polyethylene (low density)	-110 (-165)	115 (240)
	Polytetrafluoroethylene	-97 (-140)	327 (620)
	Polyethylene (high density)	-90 (-130)	137 (279)
	Polypropylene	-18 (0)	175 (347)
T_g and T_m increase with mer complexity	Nylon 6,6	57 (135)	265 (510)
	Polyester (PET)	69 (155)	265 (510)
	Polyvinyl chloride	87 (190)	212 (415)
	Polystyrene	100 (212)	240 (465)
	Polycarbonate	150 (300)	265 (510)

Degree of Polymerization

Molecular Weight



- Polymers are commercially available in sizes ranging from hundreds to millions.
- Polymers with the same constitution and configuration but having different molecular weights will have different properties.

MW affects the polymer properties significantly:

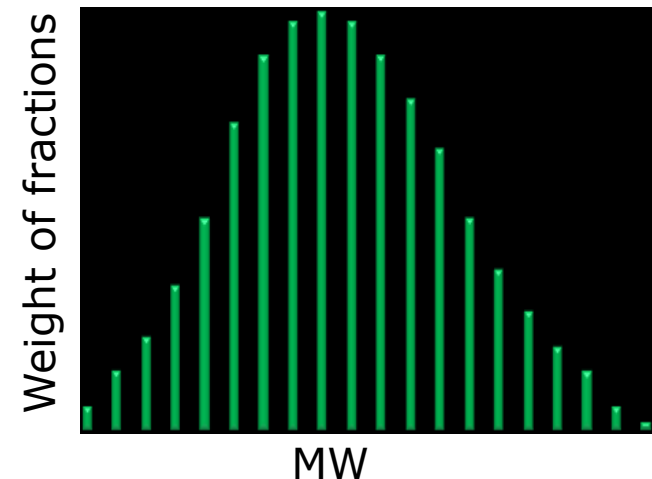
- Polystyrene (PS) with MW below 10,000 is a viscous liquid (cannot be used alone as is).
- PS with MW 10,000 or around is a brittle solid.
- PS with MW 250,000 is a hard plastic.

Molecular Weight Distribution

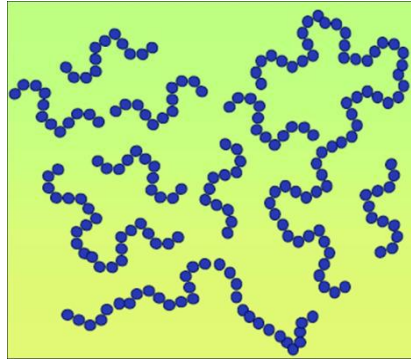
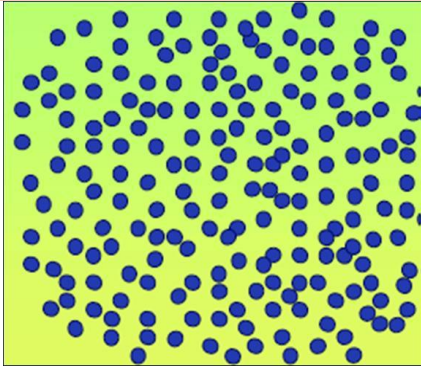
- Polymer is composed of chains of different length, consequently different weights.
- Before getting a formula to be used in calculating the average molecular weight, we must first realize that the molecular weight distribution (MWD) of polymers **is not a continuous function** of the molecular weight (M).
- Rather, only discrete values of M multiples of the monomer molecular weight (M_0).

Polymer Fractionation:

To measure the molecular weight and have the polymer distribution, the polymer sample must be fractionated. This is done by separating the sample into fractions containing narrow ranges of molecular weights.



Average Molecular Weight



300 monomers form 10 chains.
Average chain length = $300/10 = 30$

7 chains of small MW

1 chain of medium MW

2 chains of big MW

Average MW = $0.7 \times MW_s + 0.1 \times MW_m + 0.2 \times MW_b$

Note:

- Chains cannot be counted.
- Weight of each fraction and its MW are known.
- Contribution of the long chains to the properties is not considered in this example.
- The number fraction is better replaced by the weight fraction.

Indicative example

2 persons each weighs 10 kgs
 2 persons each weighs 40 kgs

$$\text{Number Average weight} = \frac{2}{4} \times 10 + \frac{2}{4} \times 40 = 0.5 \times 10 + 0.5 \times 40 = 5 + 20 = 25$$

$$\begin{aligned} \text{Weight Average weight} &= \frac{2 \times 10}{100} (10) + \frac{2 \times 40}{100} (40) \\ &= \frac{20}{100} (10) + \frac{80}{100} (40) \\ &= 0.2 \times 10 + 0.8 \times 40 = 5 + 32 = 37 \end{aligned}$$

Number Fraction

Weight Fraction

Molecular Weight

- Since not all chains in a sample of material are the same length, and so there is a distribution of molecular weights

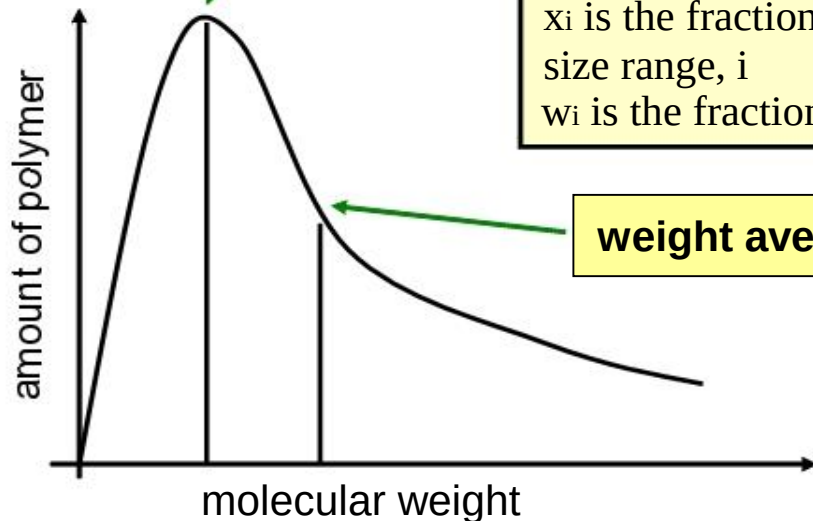
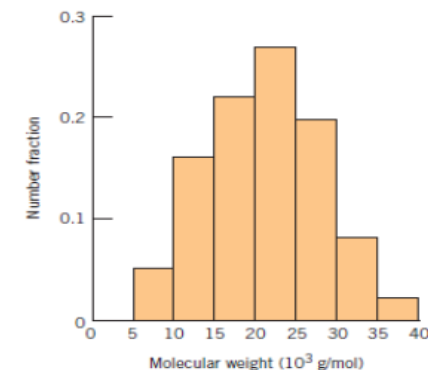
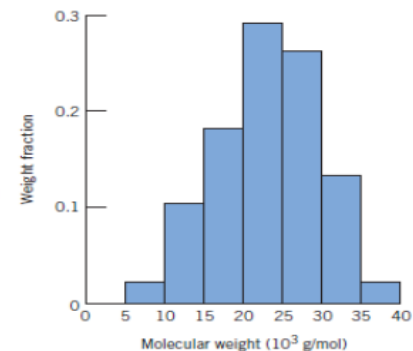
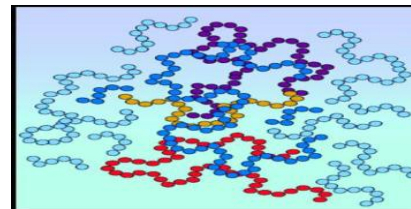
number average $M_n = \sum x_i M_i$

M_i is mean weight in size range, i
 x_i is the fraction of total number of chains in size range, i
 w_i is the fraction of total weight in size range, i

weight average $M_w = \sum w_i M_i$

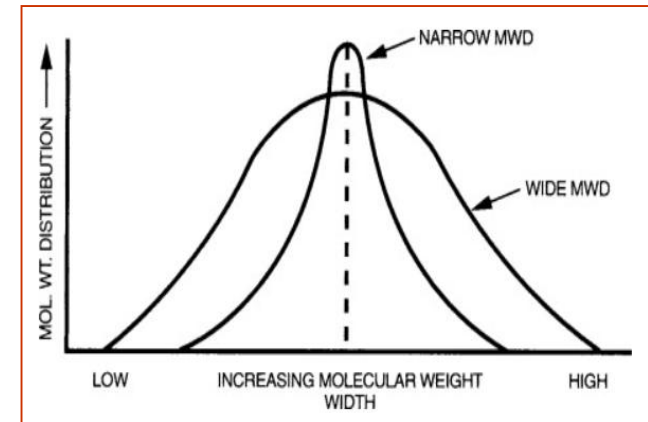
$M_w > M_n$

- If all chains were the same size, **M_w** would have been the same value as **M_n**.



Polydispersity Index (PDI)

- PDI is the ratio of the weight average molecular weight (M_w) to the number average molecular weight (M_n).
- It is always bigger than 1 as $M_w > M_n$.
- Monodisperse polymer is a polymer where $M_w = M_n$, which practically does not exist.
- High PDI indicates that MW of the polymer has a broad distribution signifying irregularity and hence low polymer performance.
- On the other hand, low PDI indicates that MW of the polymer has a narrow distribution signifying regularity and hence high polymer performance.



Degree of Polymerization

The degree of polymerization (DP or X_n) is defined as the number of monomer units in the polymer chain. It is calculated as the ratio of molecular weight of a polymer and molecular weight of the repeat unit.

Alternative way to express average polymer chain size is **degree of polymerization** which is the average number of monomer units in a chain:

number-average:

$$n_n = \frac{\overline{M}_n}{\overline{m}}$$

weight-average:

$$n_w = \frac{\overline{M}_w}{\overline{m}}$$

$\overline{m}$ is the mer molecular weight

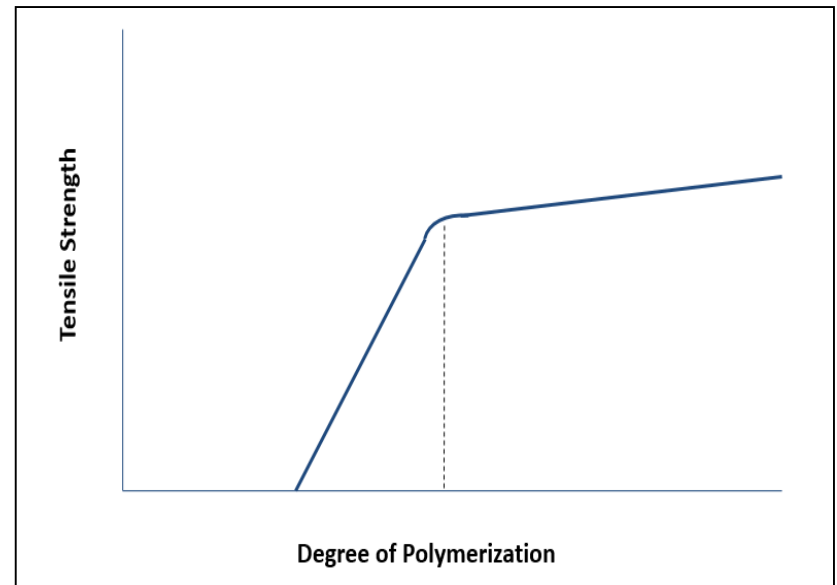
The mer molecular weight for a co-polymer can be determined by:

$$\overline{m} = \sum f_j m_j$$

f_j chain fraction of mer j
 m_j molecular weight of mer j

Effect of MW and Degree of Polymerization (DP) on Polymer Properties

- High MW enhances the mechanical properties (impact, tensile, hardness and folding strength).
- It also affects solubility, melt viscosity, solution viscosity and melting point (HOW?).
- DP represents the MW.
- Crystallinity decreases as the MW increases. (WHY?)
- Polymers reach suitable properties only after reaching a certain value of DP, i.e. MW, as seen in the opposite figure.



Effect of MW on tensile strength of cellulose acetate

Degree of Polymerization

$$\overline{DP} = \frac{N(o) - N(t)}{P(t)}$$

Where

N (o) = initial number of monomers in the feed (determined by monomer flow rate).

N(t) = number of monomers at time (t) (unreacted monomers)

P(t) = number of polymer molecules present at time(t)

Crystallizability of polymers:

- Polymers are either crystalline or amorphous (mostly both).
- Maximum crystallinity is 90%.
- Crystallizability is determined by configuration, molecular packing and chain flexibility.

Molecular Weight Calculation

Now, you can calculate the number average and weight average molecular weights for a small sample. Usually you will have information about the molecular weight distribution. The data will look something like this:

Number of Molecules

Mass of each Molecule

1	800,000
3	750,000
5	700,000
8	650,000
10	600,000
13	550,000
20	500,000
13	450,000
10	400,000
8	350,000
5	300,000
3	250,000
1	200,000

$\sum x_i M_i$
1/ 100 * 800,000
3/100 * 750,000
5/100* 700,000
8/100 * 650,000
10/100 * 600,000
13/100 * 550,000
20/100 * 500,000
13/100 * 450,000
10/100 * 400,000
8/100 * 350,000
5/100 * 300,000
3/100 * 250,000
1/100 * 200,000

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Molecular Weight Calculation

The number average molecular weight is the total weight of the sample divided by the number of molecules in the sample. What's the total weight of the sample? First find the total weight of each type (or weight) of polymer. This means that you multiply the weight of the molecule by the number of molecules of that weight. For example, there are 13 molecules with a weight of 550,000 so the total weight of molecules with a weight of 550,000 is $13 \times 550,000$ or 7,150,000. (This seems like a *really big* number, but remember size DOES matter with polymers: these are *really big* molecules.) If you do this for all the types (or weights) of molecules and add the weights together you have the total weight of the sample. In mathematical form, the total mass is $\sum N_i M_i$, where N_i is the number of molecules of weight M_i .

Molecular Weight Calculation

Number of Molecules, N_i	Mass of Each Molecule, M_i	Total Mass of Each Type of Molecule, $N_i M_i$
1	800,000	800,000
3	750,000	2,250,000
5	700,000	3,500,000
8	650,000	5,200,000
10	600,000	6,000,000
13	550,000	7,150,000
20	500,000	10,000,000
13	450,000	5,850,000
10	400,000	4,000,000
8	350,000	2,800,000
5	300,000	1,500,000
3	250,000	750,000
1	200,000	200,000
Total Mass = $\sum N_i M_i$ = 50,000,000		

Molecular Weight Calculation

To get the number average molecular weight you divide the total weight Σ of the sample by the total number of the molecules. In this case $N_i = 100$. The number average molecular weight for this sample is then $\Sigma N_i M_i / \Sigma N_i = 50,000,000 / 100 = 500,000$. Again, it's a really big number because polymers are really big molecules!

The calculation of the weight average molecular weight requires that you know the *Weight Fraction*, W_i , of each type of molecule (or the fraction of the total weight represented by each type of molecule). You don't know the weight fractions from the original data, but you can calculate them. So, what is the weight fraction? Like any fraction, it's the amount of the thing of interest divided by the total amount. For weight fraction of one type of molecule, it is the weight of that type of molecule divided by the total weight of the sample. Or, expressed in mathematical form, it is $N_i M_i / \Sigma N_i M_i$. Notice that you have already calculated both $N_i M_i$ and $\Sigma N_i M_i$ when you calculated the number average molecular weight, so you just need to do the division. The weight average molecular is then calculated by taking the sum $\Sigma W_i M_i$. When you add this information to the data table, it looks like this:

Molecular Weight Calculation

Number of Molecules	Mass of Each Molecule	Total Mass of Each Type of Molecule	Weight Fraction Type of Molecule (W_i)	
(N_i)	(M_i)	$(N_i M_i)$	$N_i M_i / \sum N_i M_i$	$(W_i M_i)$
1	800,000	800,000	0.016	12,800
3	750,000	2,250,000	0.045	33,750
5	700,000	3,500,000	0.070	49,000
8	650,000	5,200,000	0.104	67,600
10	600,000	6,000,000	0.120	72,000
13	550,000	7,150,000	0.143	78,650
20	500,000	10,000,000	0.200	100,000
13	450,000	5,850,000	0.117	52,650
10	400,000	4,000,000	0.080	32,000
8	350,000	2,800,000	0.056	19,600
5	300,000	1,500,000	0.030	9,000
3	250,000	750,000	0.015	3,750
1	200,000	200,000	0.004	800

$$N = W / MW$$

Total mass= 50,000,000

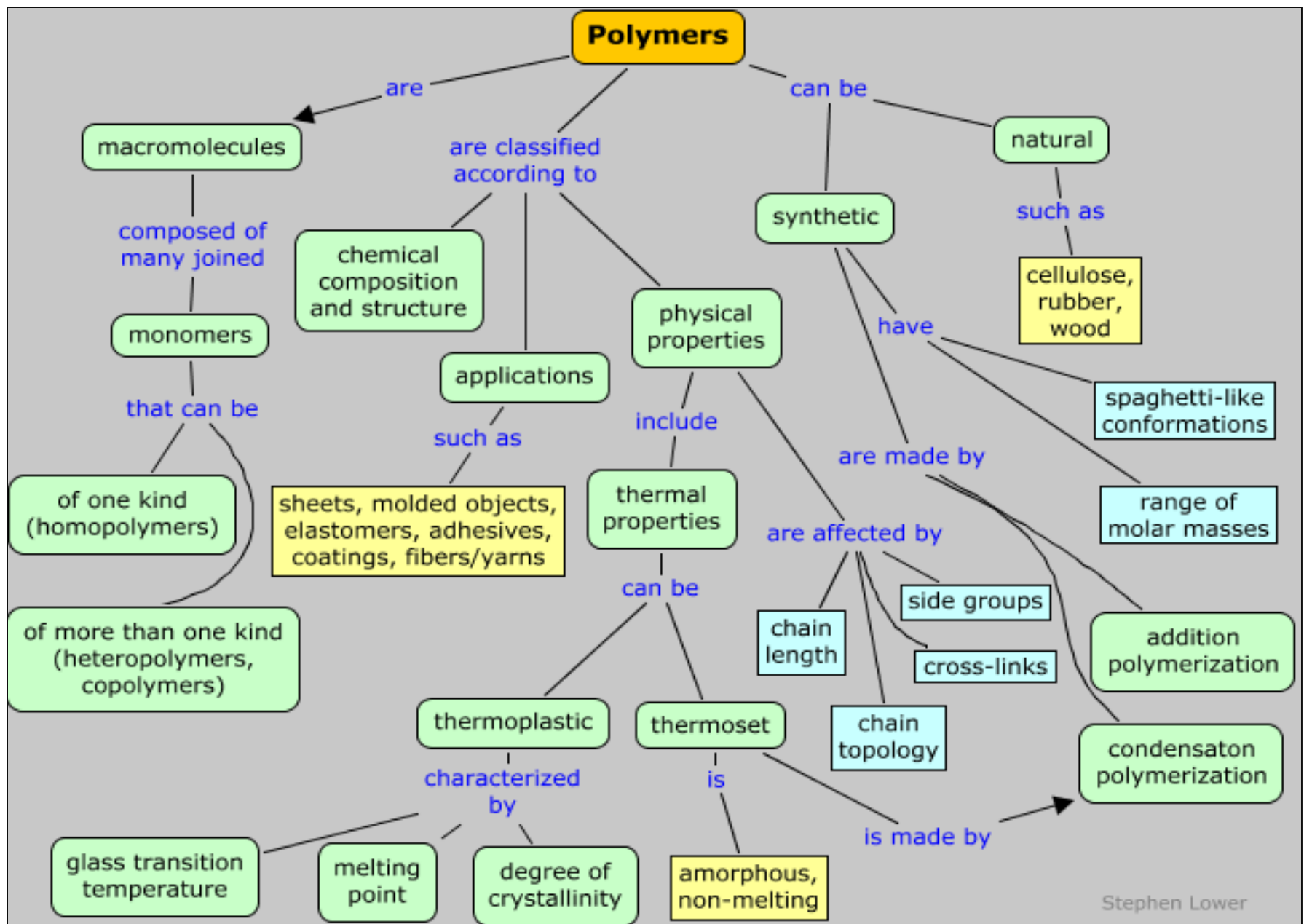
Weight Average Molecular Weight = $\sum W_i M_i$ = 531,600

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Molecular Weight Calculation

Notice that the number average and the weight average molecular are not the same. The distribution of molecular weights in a polymer sample is often described by the ratio of the weight average molecular weight to the number average molecular weight. In this case the ratio is $531,600/500,000 = 1.063$. This ratio is the Polydispersity Index (or PDI).

NOTE: The PDI must be greater than or equal to 1. When it is equal to 1, this means that the distribution is monodisperse (all the polymer chains in the MWD are of the same MW). For all real polymers the PDI is greater than 1 and can be taken as a measure of the broadness of MWD.



Stephen Lower

4/12/2025

Polymerization Techniques

POLYMERIZATION TECHNIQUES

1. Addition polymerization

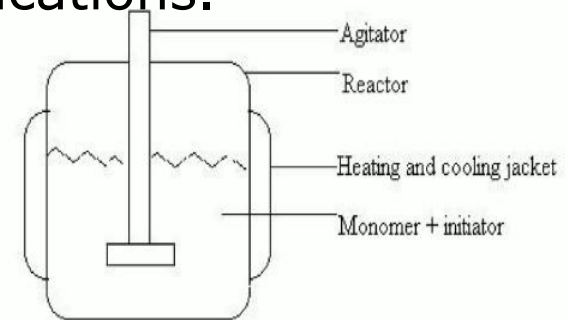
- Bulk polymerization
 - Solution polymerization
 - Suspension polymerization
 - Emulsion polymerization
- Homogenous Polymerization
- Heterogeneous Polymerization

1. Condensation polymerization

- Melt polycondensation
- Solution polycondensation

Bulk Polymerization

- ❑ Takes place in the liquid state.
- ❑ Simple, but needs thermal insulation.
- ❑ It is carried in absence of solvents.
- ❑ A typical charge might consist of a monomer-soluble initiator and perhaps a chain-transfer agent.
- ❑ The viscosity increases rapidly.
- ❑ The reaction is exothermic: once the reaction starts heating stops.
- ❑ The purest possible polymer is obtained. This may be important in electrical and optical applications.
- ❑ Very low molecular weights.



Solution Polymerization

- ❑ Carried in presence of solvent; viscosity is not very high.
- ❑ A typical charge consists of a monomer, a solvent, and an initiator or catalyst.
- ❑ The polymer solution (lacquer) can be used directly in some applications as surface coating.
- ❑ Separation of the polymer and recovery of the solvent require additional technology; obtaining highly pure polymer is difficult as traces of solvent remain in the polymer.
- ❑ Using an inert solvent in the reaction mass lowers the yield per unit volume of The reactor.



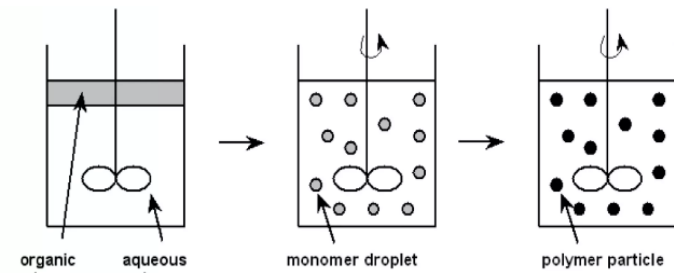
Emulsion Polymerization

- ❑ A dispersion medium is used (H₂O). Monomer (30 – 60%) is added together with an initiator. Emulsifier (synthetic detergent) is also added to lower the surface tension and increase stability.
- ❑ The resulting polymer is in the form of latex. E.g. styrene-butadiene rubber.
- ❑ High molecular weight polymer.
- ❑ Polymerization rate is very high.
- ❑ The viscosity of the reaction mass is relatively low.
- ❑ The initiator, water and emulsifier are difficult to separate.
- ❑ The latex product is often directly valuable or aids to obtain uniform compounds through master batching.

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Suspension Polymerization

- ❑ The aqueous monomer suspension is coarser than in emulsion polymerization.
- ❑ The reaction mixture consists of water insoluble monomer, monomer-soluble initiator, water, chain-transfer agent and suspending agent (stabilizer) such as gelatin.
- ❑ Each monomer droplet represents a single bulk reactor, but with dimensions small enough so that heat removal is no problem, i.e. heat is easily removed from the low-viscosity, inert suspension medium.
- ❑ The size of the product beads depends on strength of agitation.
- ❑ There is a low yield per reactor volume.
- ❑ The polymer resulting is of less purity due to the presence of the suspending agent.



Comparison between Polymerization Methods

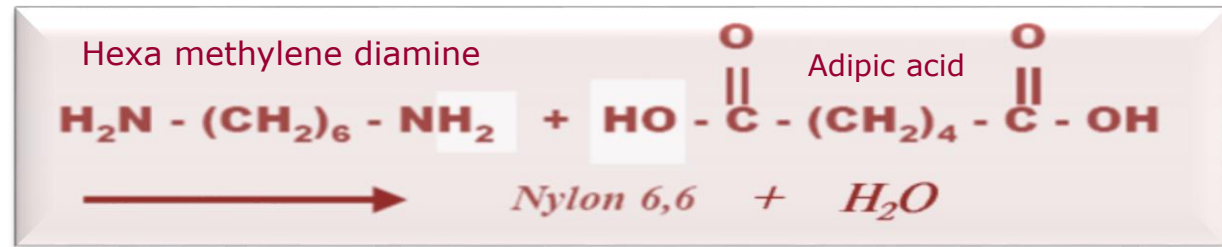
Homogenous Phase		Heterogeneous Phase	
Bulk Polymerization	Solution Polymerization	Emulsion Polymerization	Suspension Polymerization
• Reaction is highly exothermic, hard temp. control due to high rate of polymerization leading to high viscosity. • Temp in the bulk is higher than temp. at the edges, difference may be up to 20°C • Not all the monomer is converted which is difficult in separation • High purity but broad MW distribution • High conversion but low MW.	• Easier temp. control, using solvent leads to lower viscosity. • Removal of solvent and unreacted monomer is difficult. • Average MW is lower than that of bulk polymerization. • Reaction rate is slower than in bulk polymerization.	• High rate of reaction. • High average MW • Viscosity of the reaction mass is relatively low • Less conversion than in bulk polymerization • Big amount of impurities in the obtained latex	• Low rate of reaction. • Easier temp. control. • High purity. • Similar to bulk polymerization. • Average MW is lower than bulk polymerization. • Stabilization and continuous agitation are required. • Heat removal is easy due to low viscosity

Polymers Classifications

Types of Reaction

Originally, polymers were classified according to the **type of reaction** to **condensation** and **addition** polymers (and sometimes a third category called **ring-opening**).

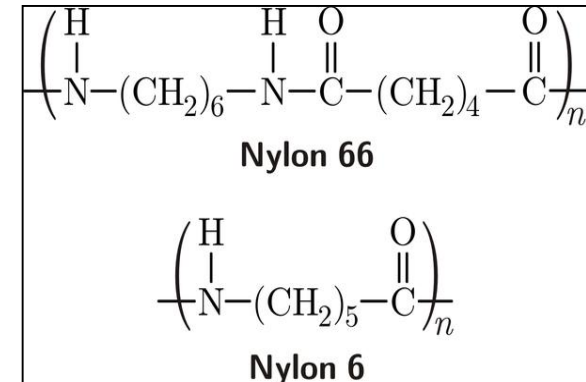
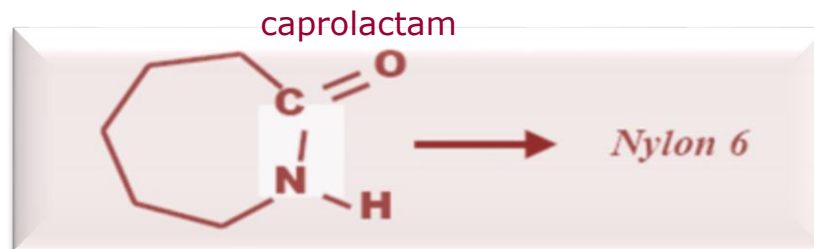
Condensation



Addition



Ring-opening



Polymers Classifications

Reaction Mechanism

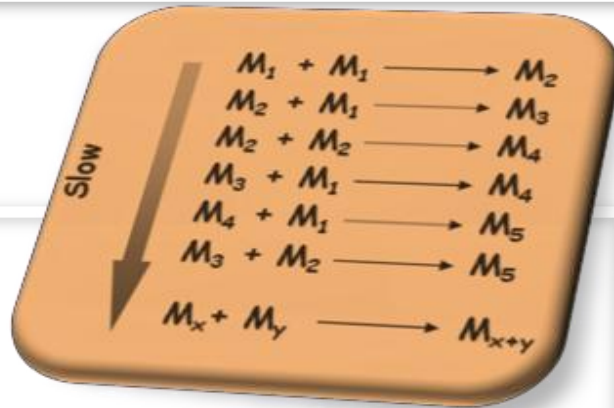
A better and more general classification is to classify the polymers to:

§Step-growth reaction mechanism

§Chain-growth reaction mechanism

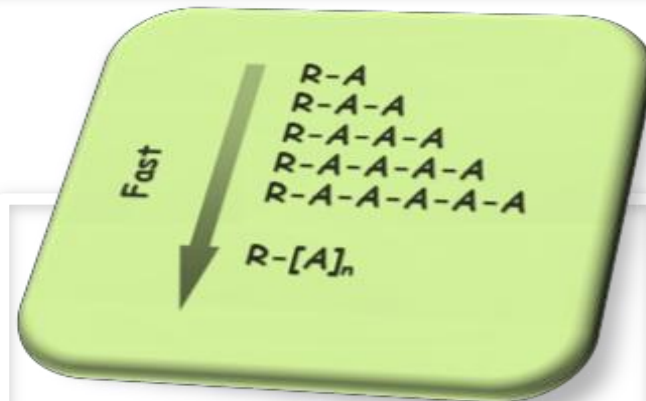
Polymers Classifications

Reaction Mechanism



Step-Growth Polymerization

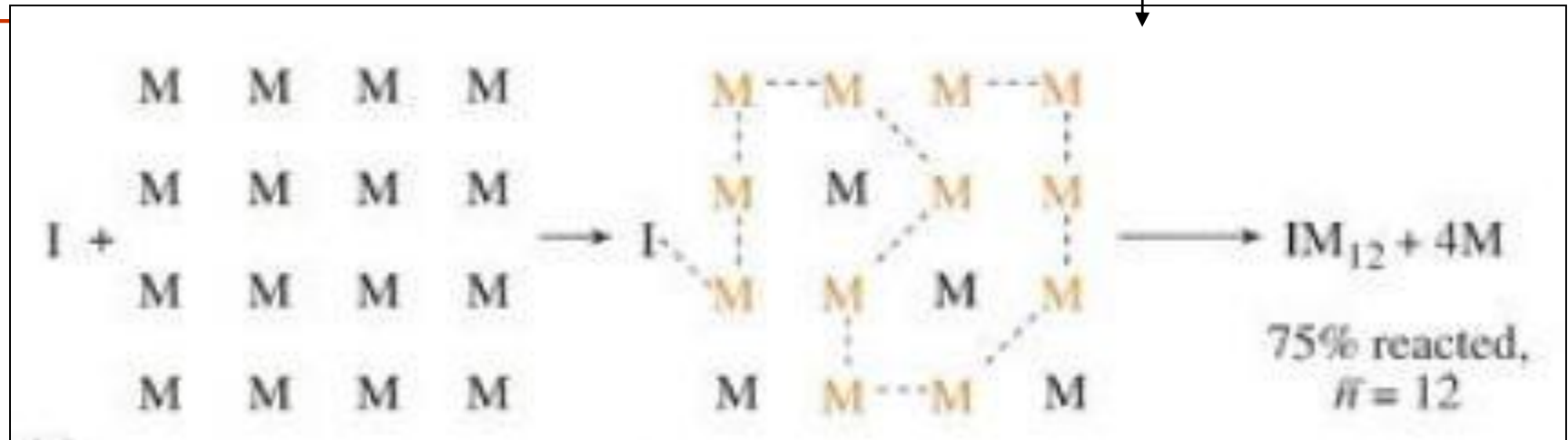
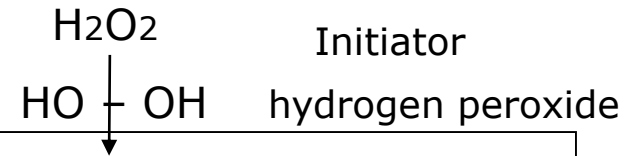
Polymerization mechanism involving the build-up of molecular weight in a stepwise fashion with random combination to form first dimers, then trimers, longer oligomers and eventually long chain polymers. Monomers involve **bi-functional** or **multifunctional** monomers.



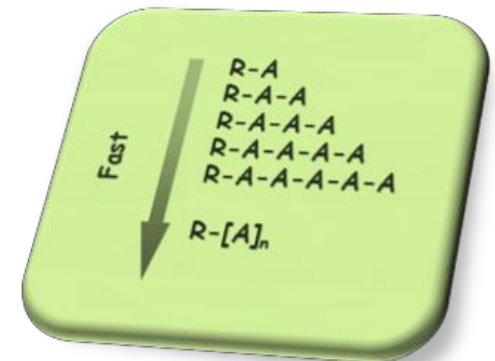
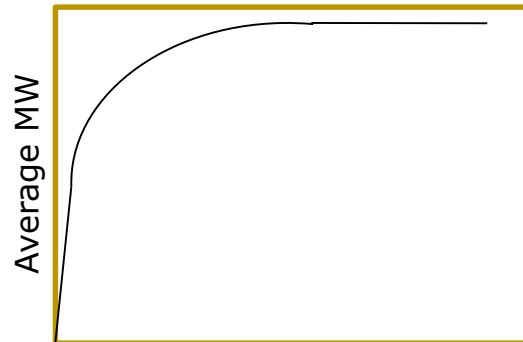
Chain-Growth Polymerization

Polymerization mechanism involving the build-up of molecular weight by adding monomer on a growing chain one at a time. Monomers involve unsaturated monomers such as **vinyl compounds**, **vinylidene compounds** and **alicyclic monomers** (cyclic aliphatic, no benzene ring)

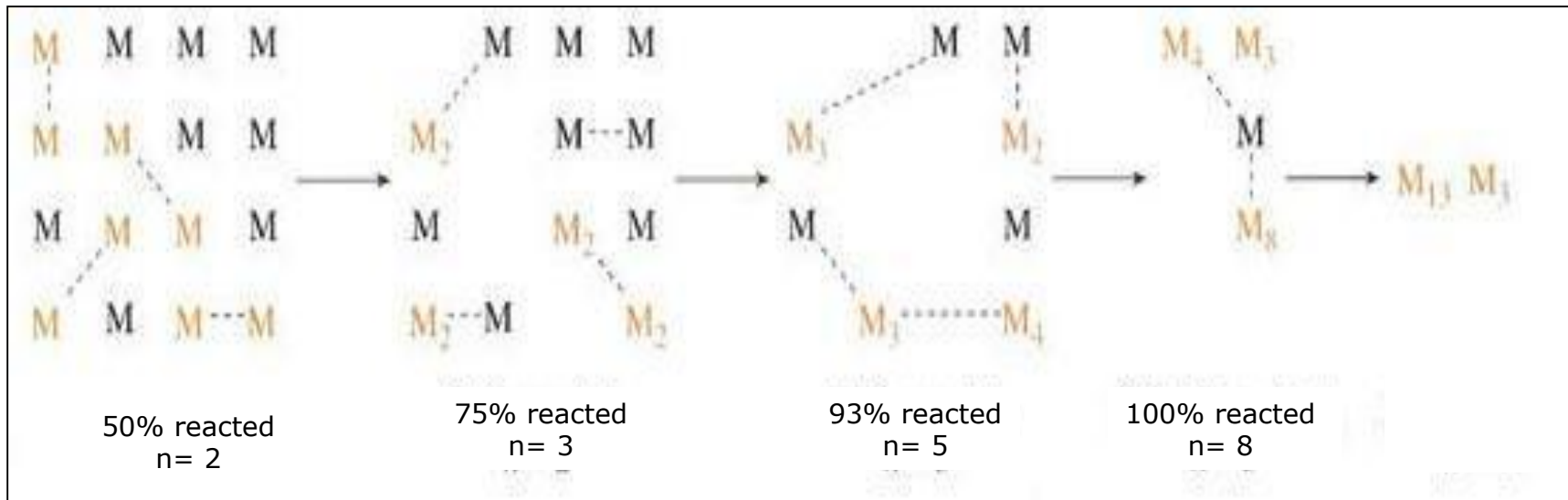
Chain-Growth Polymerization – Characteristics



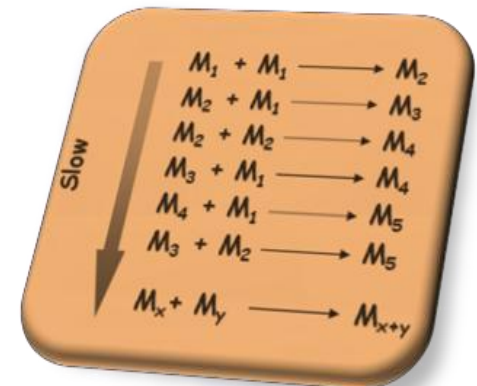
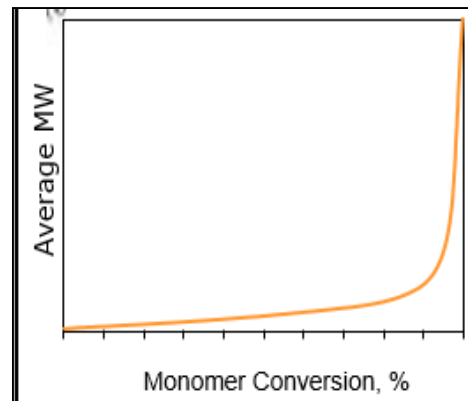
High-molecular-weight polymer is produced instantaneously at the beginning of polymerization and the molecular weight remains independent of conversion



Step-Growth Polymerization – Characteristics



Long reaction times are necessary to produce high molecular weights.
Molecular weight of polymers produced by step-growth polymerization tends to be less than those produced by chain-growth polymerization



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Polymers Classifications

Reaction Mechanism

Chain-Growth Polymerization – Examples

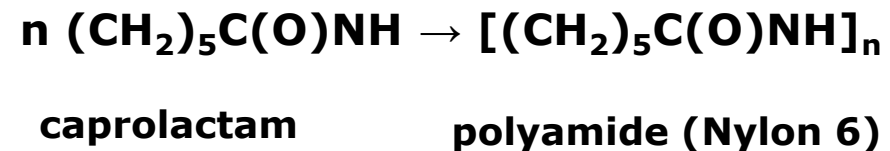
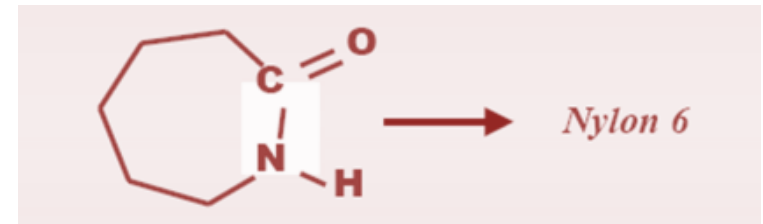
Polymer	Monomer	Repeat Unit
Poly(methyl methacrylate) (perspex)	$\text{CH}_2 = \underset{\text{CO}_2\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}$	$\text{---CH}_2\text{---}\underset{\text{CO}_2\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}\text{---}$
Poly(vinyl acetate) (adhesives)	$\text{CH}_2 = \text{CH} - \text{OCOCH}_3$	$\text{---CH}_2\text{---}\underset{\text{OCOCH}_3}{\text{CH}}\text{---}$
Poly(vinylidene chloride) (Saran fibers)	$\text{CH}_2 = \underset{\text{Cl}}{\overset{\text{Cl}}{\text{C}}}$	$\text{---CH}_2\text{---}\underset{\text{Cl}}{\overset{\text{Cl}}{\text{C}}}\text{---}$
Polytetrafluoroethylene (teflon)	$\underset{\text{F}}{\overset{\text{F}}{\text{C}}} = \underset{\text{F}}{\overset{\text{F}}{\text{C}}}$	$\text{---}\underset{\text{F}}{\overset{\text{F}}{\text{C}}}\text{---}\underset{\text{F}}{\overset{\text{F}}{\text{C}}}\text{---}$
Polyisoprene (natural rubber)	$\text{CH}_2 = \underset{\text{CH}_3}{\text{C}} - \text{CH} = \text{CH}_2$	$\text{---CH}_2\text{---}\underset{\text{CH}_3}{\text{C}} = \text{CH---CH}_2\text{---}$

Polymers Classifications

Reaction Mechanism

Chain-Growth Polymerization – Characteristics

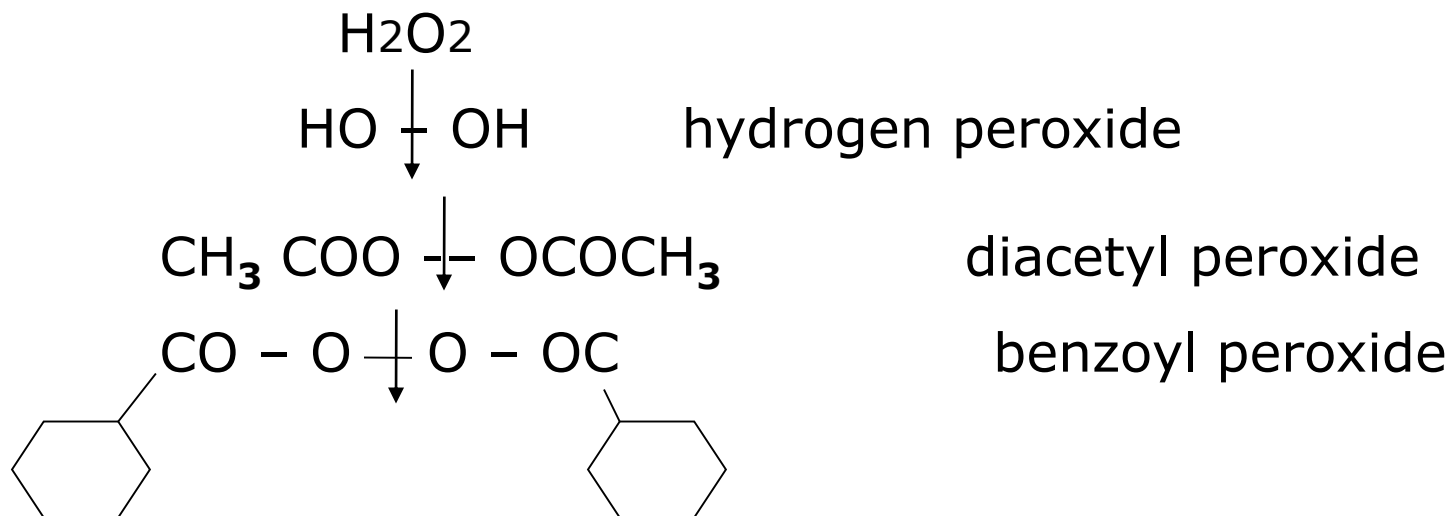
1. Monomers appropriate for chain-growth polymerizations either contain double or triple bonds or are alicyclic (compounds in which three or more atoms of the element carbon are linked together in a ring) compounds.



Chain-Growth Polymerization – Characteristics

2. Chain activity is initiated by an initiator or a catalyst. In most cases chain-growth polymerization **is not accompanied by the elimination of a small molecule** and the monomers add to the growing chains. From this the name polyaddition reactions came.

Examples of Initiators



Chain-Growth Polymerization – Characteristics

3. Chain-growth polymerizations involve:
 1. A start-up step (**initiation**).
 2. Followed by many identical chain-reaction steps (**propagation**).
 3. Finally, the active center disappears in a (**termination**) step.
- Depending on the nature of the active center, chain-growth reactions are subdivided into radical, ionic (anionic, cationic), or transition-metal mediated (coordinative) polymerizations.
- Whether a monomer polymerizes via any of these chain-growth reactions - radical, ionic, coordinative - depends on its constitution and substitution pattern. Also, external parameters like solvent, temperature, and pressure may also have an effect.

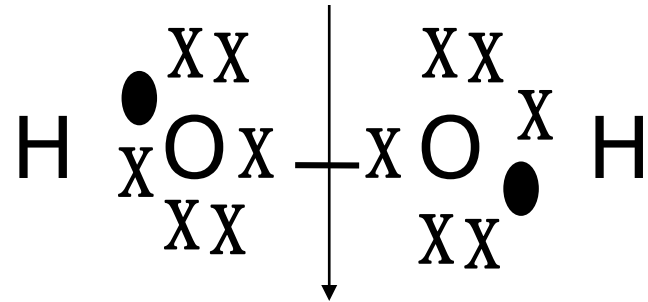
Reaction Mechanism

1. Active Centre Formation (Initiation):

H_2O_2 hydrogen peroxide initiator



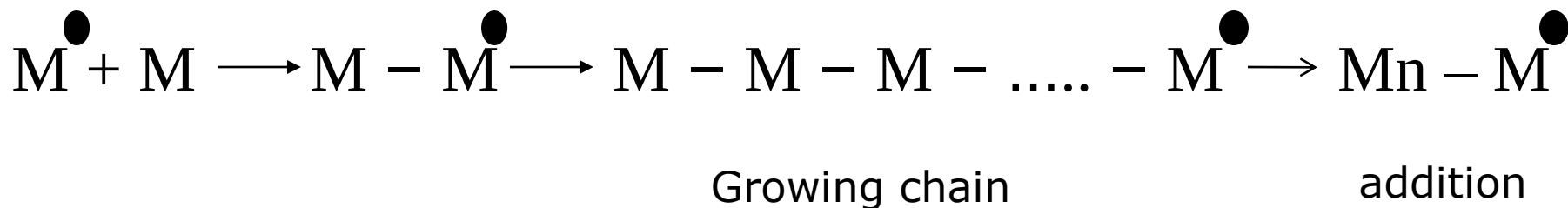
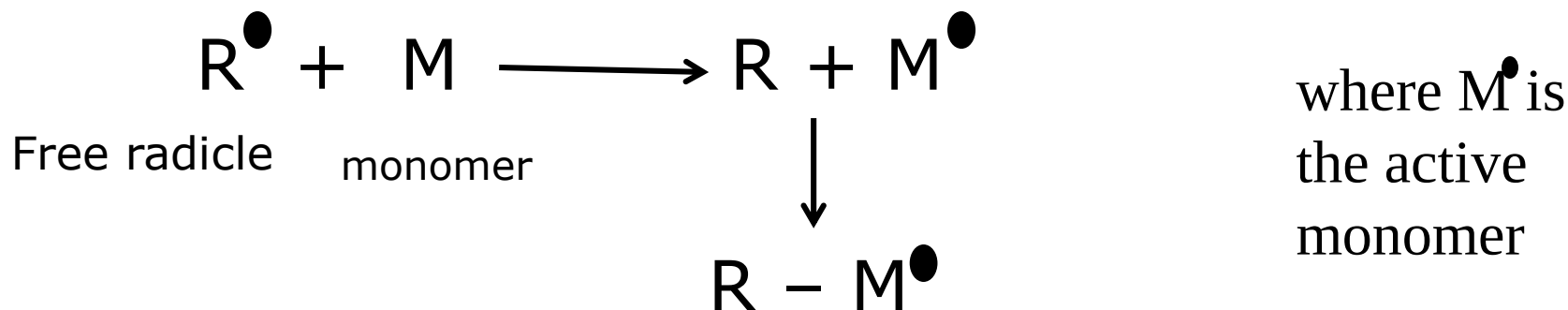
breaking of bond
by applying a
source of energy



$2 \text{HO}^\bullet$ hydroxile
free radical

Reaction Mechanism

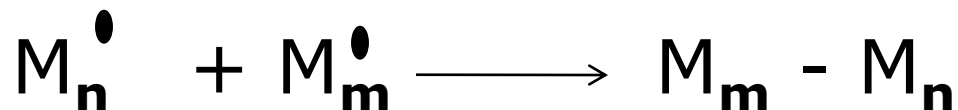
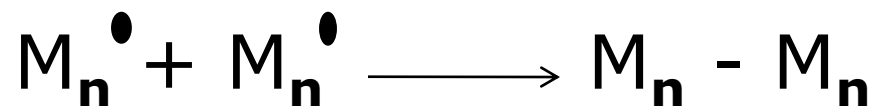
2. Propagation



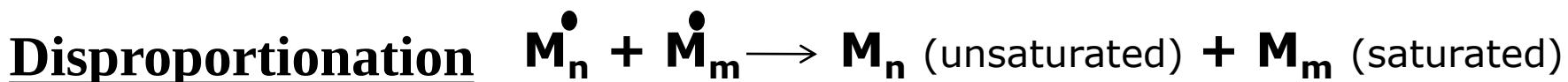
Reaction Mechanism

3. Termination

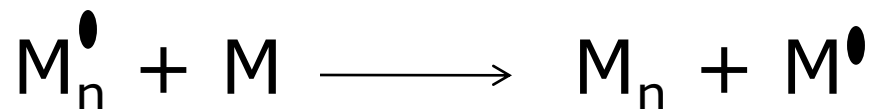
Coupling



Disproportionation

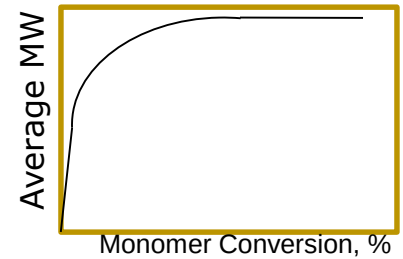


Chain Transfer

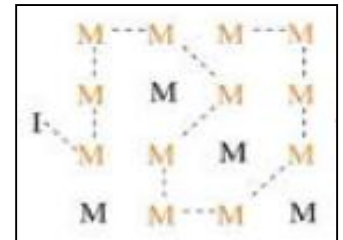


Termination takes place by one of three ways:
coupling, disproportionation or chain transfer

Chain-Growth Polymerization – Characteristics



4. (a) There are no intermediate-sized molecules in the reaction mixture— **only monomer, high-polymer, and initiator** species.
- (b) **Monomer consumption is slower** than in step-growth polymerization.
- (c) **High-molecular-weight polymer is produced instantaneously at the beginning of polymerization** and the molecular weight remains independent of conversion (VI). The only change that occurs with conversion (i.e. reaction time) is the continuous **increase in the number of polymer molecules**.



Chain-Growth Polymerization – Characteristics

5. Chain-growth polymerization is always **exothermic** with high enthalpy of polymerization (heat of reaction) in the order of 15 - 23 kcal/mol.
6. More **challenging to control** than step-growth polymerization because of the **large heat of reaction** and the **high viscosity**

Why high viscosity?



since the high molecular weight is produced from the beginning of the reaction, the viscosity increases rapidly.

Polymers Classifications

Reaction Mechanism

Step-Growth Polymerization – Characteristics

1. Monomers should have **at least 2 functional groups** for polymer to be formed ($f \geq 2$). Increasing f results in production of branches and cross-links.
2. Functional groups **react without an initiator**; reaction may be catalyzed to accelerate the reaction. In most cases (>99%), step-growth polymerization is accompanied by the **elimination of a small molecule such as water** as a byproduct. **From this the name polycondensation reactions came.**

Functionality

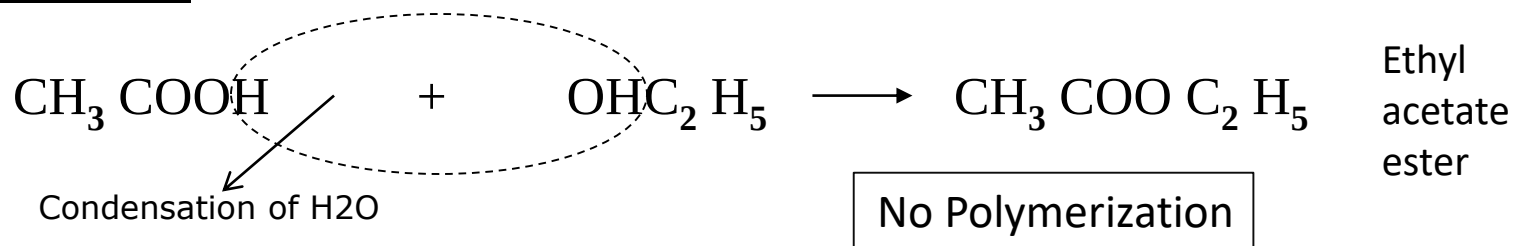
- In chemistry, **functionality** is the presence of functional groups in a molecule.
- In polymer chemistry the *functionality* of a monomer means its number of polymerizable groups, it affects the formation and the degree of crosslinking of polymers.

Functional groups

—OH	→	alcohols	;	—COOH	→	acids
—CHO	→	aldehydes	;	—NH ₂	→	amines
—SH	→	mercaptains	;	—SO ₃ H	→	sulfonic acid

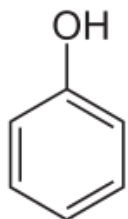
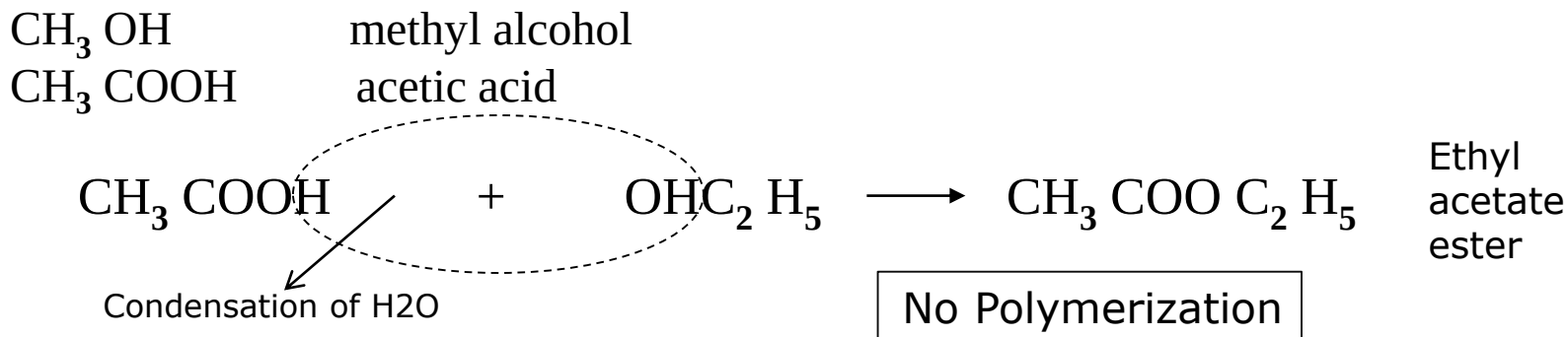
Functionality

- The functionality of a monomer is defined as the number of bonds that a monomer's repeating unit forms in a polymer with other monomers.
- In the case of a functionality of $f = 2$ a linear polymer is formed by polymerizing (a thermoplastic).
- Monomers with a functionality $f \geq 3$ lead to a branching point, which can lead to cross-linked polymers (a thermosetting polymer).
- Monofunctional monomers do not exist as such molecules lead to a chain termination.



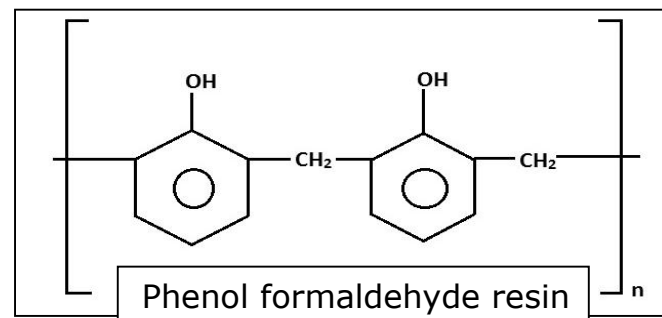
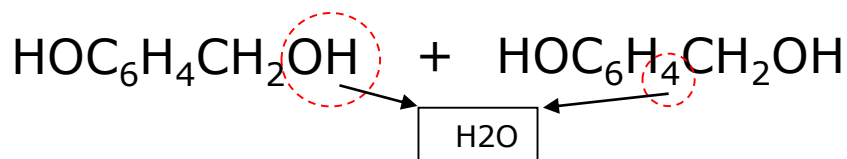
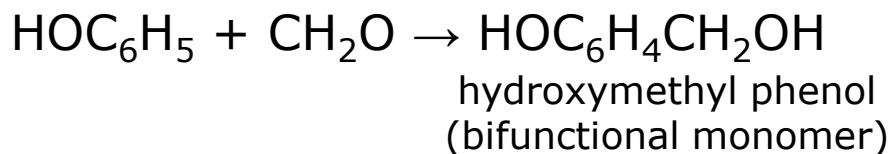
Functionality

1. Mono functional Compounds:



phenol

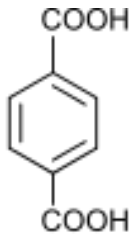
HCHO
formaldehyde



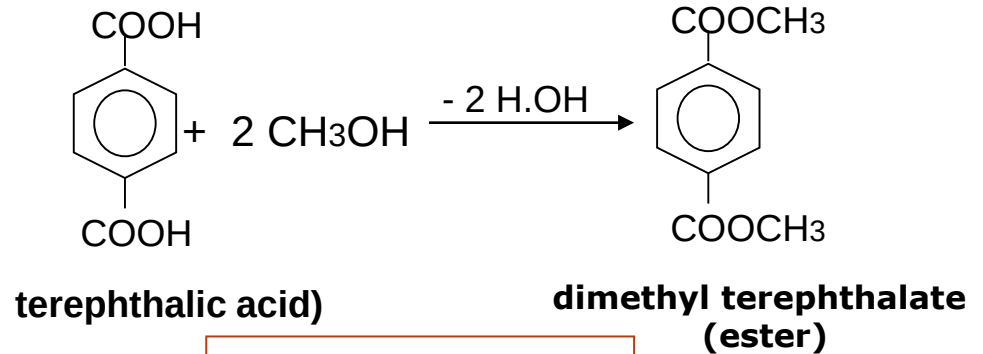
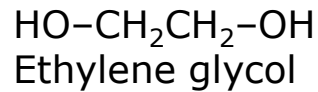
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Functionality

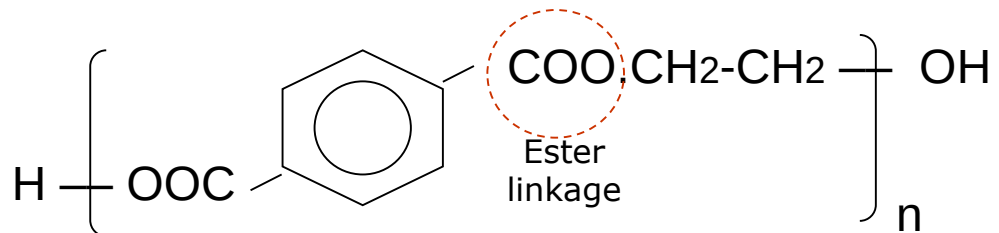
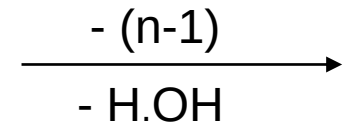
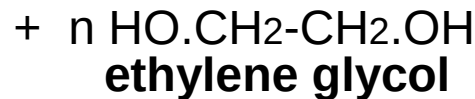
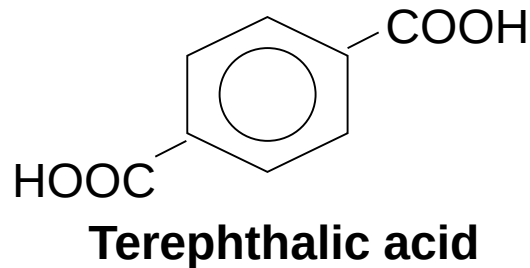
2. Bi-functional Compounds:



Terephthalic acid



No Polymerization

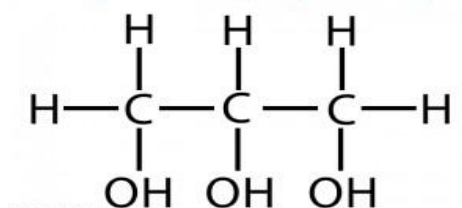


Polyethylene terephthalate, PET (polyester)

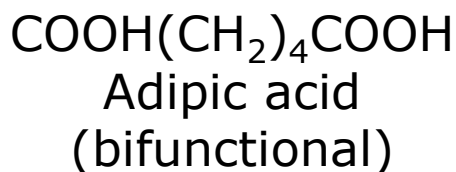
Functionality

3. Poly-functional Compounds

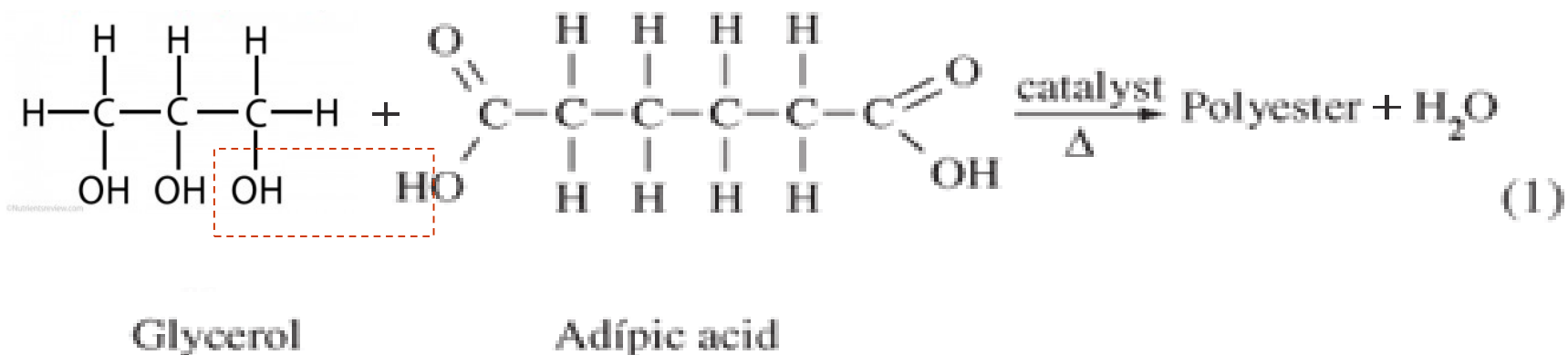
Glycerol (Glycerin)



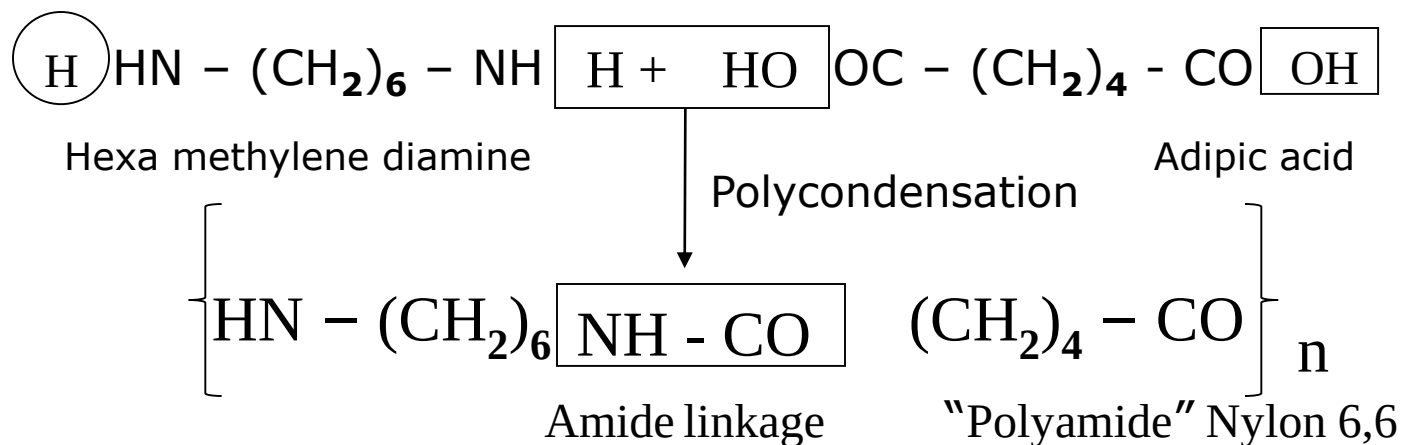
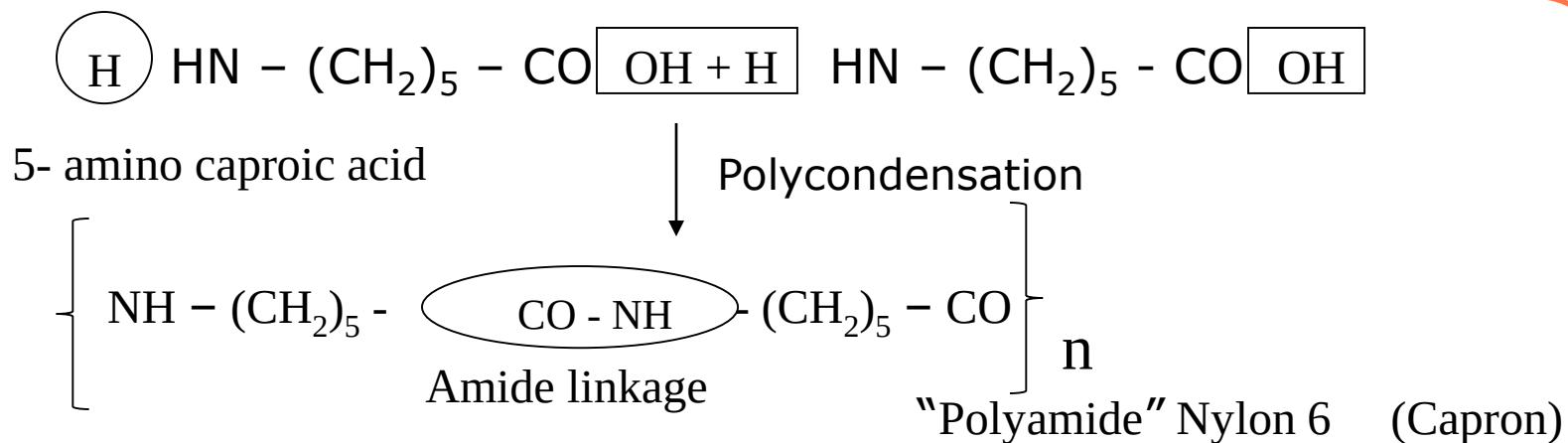
Poly-functional



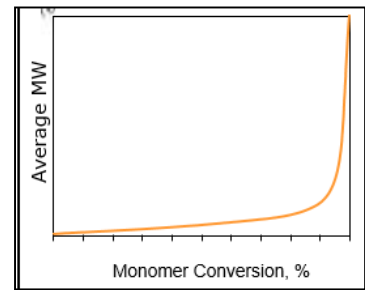
**Polymerization
results in a
branched polymer**



Step-Growth Polymerization – Examples



Step-Growth Polymerization – Characteristics



3. a) **Monomers disappear early** in reaction: at $DP_n = 10$, less than 1% monomer remains. Monomer molecules first give oligomers, which have the same reactive functional groups as monomers. Any two molecules (polymeric or monomers) in the reaction mass may react. **All chains are equally reactive regardless of MW.**

b) The average molecular weight increases steadily with reaction time; **long reaction times are necessary to produce high molecular weights.** High polymer is formed only towards the end of the reaction (when monomer conversion >99). However, the average **molecular weight of polymers produced by step-growth polymerization tends to be less** than those produced by chain-growth polymerization

(Typical S-G Mwt 40,000-60,000 versus typical C-G Mwt $10^5 - 10^6$)

Step-Growth Polymerization – Characteristics

5. Since the step-growth polymerizations are reversible (equilibrium reactions), depolymerization and recycling the polymers back to their monomers or oligomers is straight forward compared to chain-growth polymerization.
6. In most step-growth processes, the final conversion, and hence molecular weight, is limited by the reaction equilibrium. Equilibrium functional-group concentrations are affected by the concentration of the condensation byproduct. Removing the byproduct condensate from the reaction medium reduces its concentration and lowers the rate of the reverse reaction, thereby encouraging net consumption of functional groups and production of higher-molecular weight polymer.
7. Step-growth polymerizations are mildly exothermic with a low enthalpy of polymerization (heat of reaction) in the order of 2 - 6 kcal/mol